Evaluating and Extending EM-MNL Demand Estimation under Stockout Censoring in Fashion Retail

A Simulation-Based Robustness Analysis and Neurosymbolic Extension Applied to Multi-Retailer Sales Data

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Abstract

Stockout censoring biases sales data by shifting unrecorded demand to substitute products. While the Expectation-Maximisation Multinomial Logit (EM-MNL) model can recover this hidden demand, its reliability in realistic fashion retail environments remains largely unproven due to past reliance on single-retailer, assumption-compliant datasets. This thesis evaluates EM-MNL using simulations and multi-retailer fashion data. Under ideal conditions, EM-MNL accurately recovers demand, outperforming naive baselines, Poisson regression, and neural networks. A robustness analysis identifies thresholds for substitution correlation and arrival overdispersion where this advantage fades, giving practical application criteria that are validated across twelve retail stores. Finally, the thesis introduces a neurosymbolic extension of EM-MNL that integrates style-based substitution knowledge as a soft constraint. Correct rules enhance demand recovery, particularly when dealing with sparse data, whereas incorrect rules degrade performance. Disabling the rule reverts the system to the original estimator.

Keywords

Demand estimation, EM-MNL, Fashion retail, Neurosymbolic AI

GitLab Repository

https://gitlab.com/wair-thesis-max/thesis_project

1 Introduction

1.1 Background and Problem Definition

Demand forecasting is a fundamental component of retail operations, influencing inventory management, replenishment planning, and supply chain optimisation [7]. Accurate forecasts enable retailers to minimise stockouts, reduce excess inventory, and improve customer satisfaction [17]. However, a critical challenge in retail demand forecasting is that observed sales data often provide a biased and incomplete representation of true customer demand [20].

A key cause of this bias is stockouts and substitution behaviour [23]. When a product is unavailable, customers may substitute to an alternative product or choose not to purchase at all. In such cases, the observed sales of the unavailable product underestimate its true demand, resulting in censored observations. As a result of this, standard forecasting models trained on raw sales data systematically underestimate demand for frequently unavailable products and overestimate demand for products often picked as substitutes. This inaccuracy propagates through forecasting pipelines, resulting in suboptimal inventory decisions, reduced operational efficiency, and increased costs [3].

To address this issue, various methods have been developed to recover primary demand. Early research established the systematic underestimation that stockouts create in standard sales estimates [23]. To correct this, simpler heuristic methods adjust the data by proportionally scaling up sales based on how long a product was actually on the shelf [3]. More advanced choice-based approaches instead model how customers substitute one product for another. These methods use discrete-choice models, typically solved with the Expectation-Maximization (EM) algorithm, to reconstruct the missing data for customers who either bought a substitute or left without purchasing [3, 20]. Once primary demand is recovered, these corrected figures can be used to train standard machine learning forecasting models like Random Forest and SVR [14]. One prominent approach is the Expectation-Maximization Multinomial Logit (EM-MNL) model, which estimates customer choice behaviour and aims to reconstruct primary demand using only observed sales and product availability data [20]. This approach accounts for substitution effects and corrects demand estimates for censoring bias.

Prior and recent research has demonstrated the effectiveness of EM-based demand recovery [14, 20, 21], yet these studies typically examine limited subsets of retail conditions in isolation and rely on data from a single retailer [6, 14, 17, 20]. The conditions under which EM-MNL estimates remain reliable are therefore poorly characterised. This thesis addresses that gap on two fronts. First, it uses controlled simulation with known ground truth to analyse estimator performance across a broader range of retail environments (substitution strength between products and customer arrival dispersion), identifying the boundaries within which EM-MNL remains reliable. Second, it applies the model to real fashion retail data provided by WAIR, an Amsterdam-based platform that aggregates sales across multiple retailers, testing whether the estimator generalises beyond a single chain.

Beyond evaluation, the thesis contributes a simple neurosymbolic augmentation of EM-MNL. Fashion retail exhibits well-understood structural relationships, such as products within a style family sharing similar baseline appeal, that a purely data-driven estimator cannot infer from censored sales and that existing recovery methods leave unused. The augmentation encodes a relationship as an explicit symbolic constraint, added to the estimation objective as a soft, weighted penalty, and tests whether this more interpretable method improves robustness without sacrificing the recovery behaviour of the core EM-MNL framework.

1.2 Research Questions

The central research question guiding this thesis is:

> **RQ:** To what extent can EM-MNL and a simple neurosymbolic extension of EM-MNL recover true primary demand from retail sales data under stockout censoring, and which approach provides greater accuracy, robustness, and interpretability in fashion retail environments?

- **SQ1 (Estimation Accuracy):** How accurately does the EM-MNL estimator recover primary demand and reproduce observable sales patterns in simulated retail environments with known ground truth, relative to a proportional uplift baseline?
- **SQ2 (Robustness and Actionable Limits):** How does EM-MNL estimation accuracy and distributional validity deteriorate when key modelling assumptions (IIA and Poisson arrivals) are progressively violated, and at what point does the estimator become no more reliable than a simple baseline?
- **SQ3 (Real-World Application):** When EM-MNL is applied to real fashion-retail sales data, how much primary demand does it recover beyond censored sales, and to what extent can its estimates be validated without ground-truth demand?
- **SQ4 (Neurosymbolic Extension):** To what extent can incorporating simple symbolic retail constraints into EM-MNL improve the robustness or interpretability of demand estimation in real-world fashion retail data?

2 Related Work

This section reviews the literature and provides context for this study. The review is organised into three thematic areas: demand censoring and choice-based estimation, with the EM-MNL framework as the state of the art; simulation evidence and model limitations in fashion retail; and neurosymbolic approaches to demand modelling. The section closes with a synthesis of the research gaps that motivate this thesis.

2.1 Demand Censoring and Choice-Based Estimation

A fundamental challenge in retail analytics is that observed sales do not necessarily reflect true customer demand. When products stock out, customers either substitute toward an alternative or abandon the purchase, so demand for the stocked-out item is underestimated while sales of available substitutes are inflated by recaptured demand. The result is a set of *censored* observations, widely referred to as demand censoring [3, 20, 23].

The consequences of ignoring censoring are well documented. Sales under stockouts are systematically biased predictors of underlying demand [23], and failing to account for substitution and lost sales propagates errors into inventory policies, lowering fill rates and raising costs relative to models estimated on corrected demand [16]; the bias worsens as stockouts grow more frequent [17]. These findings motivate correcting for censoring before any forecasting or inventory optimisation step.

A complete correction must address two forms of censoring: spilled demand, where an unavailable first choice goes unrecorded, and recaptured demand, where a substitute purchase is recorded but attributed to the wrong product. Proportional uplift heuristics address spill but not recapture, leaving the double-counting problem intact [20]. Choice-based demand estimation resolves both by modelling each arriving customer as selecting from the available assortment under a discrete choice model, jointly recovering substitution parameters and primary demand from censored transactions.

Kök and Fisher [11] develop an early EM-based approach that jointly estimates demand and substitution probabilities, but it requires individual-level store-visit data that is rarely available in practice and a computationally intensive non-linear M-step.

The most widely applied framework is the EM-MNL model of Vulcano et al. [20], which combines a Multinomial Logit (MNL) choice model with a non-homogeneous Poisson arrival process. Customers arrive at a period-specific rate, and each selects from the products available in that period according to a vector of product preference weights, competing against a no-purchase option. The key insight is to reframe estimation in terms of *primary demand* (the demand a product would have received had all products been continuously available) and to treat observed sales as incomplete realisations of it. The EM algorithm then alternates between an E-step that imputes expected lost sales for unavailable products and an M-step that updates the preference weights in closed form, converging to a stationary point of the incomplete-data log-likelihood [20]. The formal choice probabilities and update equations are given in Appendix B.

The model cannot determine the absolute scale of the preference weights from data alone, since scaling all weights and arrival rates by a common constant leaves the likelihood unchanged; the estimates therefore require an external anchor. Vulcano et al. [20] supply the retailer's market share (the fraction of arriving customers who purchase under full assortment) as an exogenous input. Their sensitivity analysis shows that mis-specifying this share by ±20% distorts the estimated preference weights but leaves the primary demand estimates largely unaffected [20], which matters because market share is rarely known with precision.

Vulcano et al. [21] provide empirical evidence on airline booking data, with downstream revenue management gains of 1.4–5.3%. Conlon and Mortimer [3] develop a closely related EM algorithm for periodic-review inventory, but its E-step becomes prohibitive when many products stock out simultaneously, requiring exponentially many parameters in the joint stockout state space. The Vulcano et al. [20] formulation sidesteps this by imposing MNL structure on choice probabilities, giving a fixed-dimensional parameter vector and closed-form E-step expressions regardless of stockout configuration. Subsequent work treats it as the canonical benchmark: Abdallah and Vulcano [1] characterise it as state of the art with an order-of-magnitude speed-up over direct MLE, while Subramanian and Harsha [15] position their MIP-based alternative as designed to dominate EM-MNL. EM-MNL has since been applied to fashion-retail demand reconstruction, outperforming sales-weighted averaging baselines [14]. This thesis nonetheless adopts EM-MNL rather than the MIP-based estimator of Subramanian and Harsha [15]. That estimator extends Berkson's log-ratio method rather than the EM-MNL likelihood and, as its authors note, requires grouped data and a heuristic correction wherever zero sales are frequent. Fashion retail at the SKU level is exactly this regime: sales per SKU are extremely sparse, with most SKU-period cells recording no sale (Appendix A), so this zero-sales correction would be in near-constant use, whereas the closed-form EM E-step accommodates such sparsity directly and without an external MIP solver.

More recent work applies alternative data-driven methods to censored fashion demand. Sousa et al. [14] benchmark Support Vector Regression and Random Forest on censored fashion data and find that machine learning can partially correct for censoring when covariates (price, category, colour) are informative. However, these methods do not model the substitution mechanism and so cannot reconstruct the substitution matrix or provide per-product lost-sales estimates, outputs that are central to inventory planning. EM-MNL therefore remains the primary method when interpretable, structurally grounded demand recovery is required and individual-level choice data are unavailable. Nevertheless, its reliance on the IIA assumption, Poisson arrivals, and EM updates that converge only to local optima motivates richer alternatives. Nested and random-coefficient choice models improve behavioural realism at the cost of substantially greater data and computational requirements, while machine-learning approaches sacrifice explicit substitution recovery. In practice, EM-MNL persists because it offers a tractable compromise between realism, interpretability, and applicability to sparse aggregate retail data.

2.2 Simulation Evidence, Model Limitations, and Validation Scope

Because true primary demand is unobservable in real retail data, simulation-based recovery experiments play a central role in validating structural demand estimators. Monte Carlo experiments generate synthetic environments in which arrivals, substitution, and availability are fully specified, enabling direct evaluation of estimator bias, variance, and consistency against known ground truth [3, 20].

The original EM-MNL paper [20] conducts an extensive Monte Carlo study. Estimation accuracy improves substantially with data volume, with most preference weight biases falling below 10% once approximately 50 or more periods are available, and with market potential, since higher market share implies more purchases per period and therefore more informative data on each product's attractiveness. When assortments change rarely, there is insufficient variation in the availability structure to identify substitution effects, and the model provides little advantage over the naïve baseline. These findings establish the practical conditions under which EM-MNL can be expected to perform reliably.

A consistent limitation is that these experiments generate data under the model's own assumptions: Poisson arrivals and IIA substitution. The estimator is therefore evaluated mainly where it is theoretically guaranteed to recover the truth. Systematic evaluation under progressively misspecified data-generating processes has not been conducted, leaving practitioners without evidence-based guidance on when to distrust EM-MNL outputs.

The MNL model relies on the Independence of Irrelevant Alternatives (IIA) assumption, under which a stocked-out product's customers purchase the remaining available items strictly in proportion to their popularity, regardless of how similar the products actually are. Fashion retail has well-documented structural features that violate this systematically, such as switching within brands [12].

Fashion assortments are organised by style, colour, and size. A customer unable to find their size substitutes preferentially toward the same or similar style in an available size rather than uniformly across the entire assortment [11], so size-specific stockouts produce clustered substitution within style families rather than the proportional redistribution IIA implies. Nested Logit and random-coefficient MNL extensions permit correlated preferences within product groups [3], but require substantially more data to identify the additional parameters and lose the closed-form M-step that makes basic EM-MNL computationally efficient.

The Poisson arrival assumption presents a separate limitation. The Vulcano et al. [20] model allows a non-homogeneous Poisson process with period-specific rates λ_t, accommodating systematic day-of-week or seasonal variation, but imposes equality of mean and variance within each period. In fashion retail, demand is punctuated by discontinuous events such as collection launches, promotions, and end-of-season clearance, which produce over-dispersed arrival counts whose variance substantially exceeds the mean [14]. This extra variance cannot be absorbed by adjusting λ_t alone and calls for a model such as the Negative Binomial; retaining Poisson leads to overestimated arrival rates and distorted lost-demand calculations.

Despite recognition of these limitations [3, 20], their quantitative impact in fashion retail has not been systematically tested: studies either acknowledge violations qualitatively or test them individually in non-fashion domains. This thesis fills that gap with a simulation design that progressively introduces IIA violations (via Nested Logit data-generation) and arrival overdispersion (via Negative Binomial processes), measuring the degradation in estimation accuracy and the threshold at which EM-MNL no longer outperforms the naïve proportional baseline.

A related gap is the censoring regime over which EM-MNL has been validated. The core experiments in Vulcano et al. [20] use availability probabilities of 0.8, 0.7, and 0.5 per period (stockout rates of 20%, 30%, and 50%), with a single higher-censoring case reported but not replicated. At very high stockout rates the few in-stock periods carry almost all identifying information, so informative availability variation collapses, which may fundamentally change how well EM-MNL identifies substitution effects.

Real-world validation is similarly constrained. Most studies apply EM-MNL or related methods to a single retailer [3, 14, 20], so it is unclear whether the estimated substitution structures and censoring corrections generalise across retail formats, price points, and customer segments. Multi-retailer datasets provide greater variation in assortment structure and stockout patterns, enabling more robust assessment of where the model's assumptions hold and where they break down.

2.3 Neurosymbolic Approaches in Operations and Retail

Neurosymbolic AI combines statistical learning with explicit symbolic knowledge, such as logical rules or domain constraints, to produce models that are both data-driven and interpretable [5]. It is most useful when data are limited but domain knowledge is strong, and has been applied in areas such as supply chain management [10] and medical text extraction [13].

A common way to add a rule to such a model is to turn it into a soft penalty on the loss function, scaled by a weight that controls how strongly the rule is enforced. This is the idea behind semantic loss [24], semantic-based regularisation [4], and Logic Tensor Networks [2]. The weight is set by hand, and because the penalty is soft, strong data can override the rule, unlike a hard constraint.

In retail, assortment structure is a natural rule. Products in the same style family are close substitutes, so a customer who cannot find their size usually switches within the style rather than across categories [11, 12]. Standard MNL estimation ignores this, and the E-step of EM-MNL can therefore assign lost demand to implausible substitutes. Adding a soft penalty that keeps same-style products similar lets the data and the rule decide together; because the EM-MNL updates are not differentiable, this is done with a gradient-trained version of the model that reduces to EM-MNL when the rule is switched off (Section 3.8). To the best of the author's knowledge, current literature has not reformulated stockout-censored EM-MNL as a differentiable optimisation problem that permits symbolic constraints to be incorporated through gradient-based regularisation. Existing work either preserves classical EM estimation for censored demand recovery, proposes alternative optimisation procedures without symbolic structure, or introduces symbolic regularisation in unrelated domains. Therefore, evaluating both the advantages of a correctly specified rule and the penalties of misspecification constitutes a primary contribution of this thesis.

2.4 Research Gap

The literature reviewed above establishes that EM-MNL is the leading method for recovering primary demand from censored retail sales data and that it consistently outperforms naïve and exponential-smoothing baselines under the conditions for which it was designed. However, several interconnected gaps limit practitioners' ability to deploy the method with confidence in fashion retail settings.

Validated censoring range. The core simulation evidence for EM-MNL covers stockout rates between 20% and 50% [20]. Fashion retail routinely operates at rates exceeding 70%, and no replicated evidence exists for this high-censoring regime.

Assumption-violation characterisation. Evaluations are conducted under data-generating processes that match the model's own assumptions. The quantitative impact of IIA violations and Poisson overdispersion (both common in fashion retail) on estimation accuracy and the relative advantage over the naïve baseline has not been characterised.

Multi-retailer generalisability. Existing real-world applications use single-retailer datasets [3, 14, 20], leaving the cross-retailer generalisability of EM-MNL demand estimates an open question.

Neurosymbolic integration. While symbolic constraints from assortment structure are a natural fit for EM-MNL, no prior work has, to the best of the author's knowledge, evaluated a differentiable reformulation of stockout-censored EM-MNL that incorporates such constraints as soft, weighted penalties while preserving the original estimator as the limiting case.

This thesis addresses all four gaps. It extends the simulation validation to high-censoring regimes and model-misspecified data-generating processes, applies EM-MNL to a multi-retailer fashion dataset, and implements and evaluates a differentiable, rule-constrained reformulation of EM-MNL. Together, these contributions provide the first systematic evidence base for practitioners deciding whether EM-MNL demand estimation is appropriate for a given fashion retail dataset.

3 Methodology

3.1 Research Design

The four research sub-questions defined in Section 1.2 map onto the four research gaps identified in Section 2.4. The methodology follows the order: SQ1 establishes baseline performance under the conditions

for which the estimator was designed; SQ2 identifies the conditions under which those assumptions break down; SQ3 applies the estimator only where the SQ1–SQ2 evidence supports it; and SQ4 augments the estimator with structural constraints derived from the same data.

SQ1 establishes a simulation-based performance baseline under ideal conditions, replicating and extending the controlled Monte Carlo experiments of Vulcano et al. [20]. SQ2 applies the same estimator to data generated under progressively misspecified data-generating processes (DGPs), identifying the thresholds at which EM-MNL no longer outperforms a simple naïve heuristic. SQ3 and SQ4 build on SQ2 by applying and extending the model to real-world fashion retail data.

3.2 Choice-Based Demand Model

The choice-based demand model follows Vulcano et al. [20] and is given in full in Appendix B; this subsection only fixes the notation used in the rest of the methodology. Let $t \in \{1, \ldots, T\}$ index time periods and $j \in \{1, \ldots, J\}$ index products, where T and J are the numbers of periods and products. Let $S_t \subseteq \{1, \ldots, J\}$ denote the set of products available in period t, and let $a_{jt} = \mathbf{1}[j \in S_t]$ be the corresponding binary availability indicator, equal to 1 when product j is available in period t and 0 otherwise. Y_{jt} represent observed sales and D_{jt} represents the latent primary demand, the sales that would have been recorded for product j in period t had all products been continuously available, which the EM algorithm treats as the complete data. The Poisson-MNL likelihood is identified only up to scale (see Section 2); the exogenous market share parameter s, representing the expected purchase probability under full availability, fixes this scale via the outside-good (no-purchase) ratio:

$$r = \frac{1 - s}{s}. \tag{1}$$

3.3 Baseline Models

All simulation and real-world analyses are benchmarked against the naïve proportional uplift estimator of Vulcano et al. [20]. The estimator approximates the underlying demand rate for product j from its in-stock periods:

$$\hat{d}_j = \frac{\sum_{t=1}^{T} Y_{jt}}{\sum_{t=1}^{T} a_{jt}}, \tag{2}$$

where $\hat{d}_j$ is the estimated per-period demand rate for product j. The estimator applies this flat rate to all stockout periods to approximate lost demand. As Vulcano et al. [20] demonstrate, this estimator systematically overestimates primary demand for products that frequently serve as substitutes (because recaptured demand inflates their observed sales) and underestimates demand for products that are often stocked out. The naïve estimator acts as a lower-bound benchmark: any additional modelling complexity in EM-MNL must be justified by a measurable accuracy gain over naïve. To strengthen the evaluation, two machine learning (ML) baselines are introduced alongside the naïve heuristic: a regularised Poisson regression, the standard generalised linear model for count-valued sales, and a multilayer perceptron (MLP), both trained directly on the censored observed sales (Section 3.5.1). The ML baselines reflect the performance of standard data-driven approaches, one linear and one non-linear, that do not explicitly model customer choice behaviour. Together, these baselines define a range of benchmark performance against which EM-MNL can be assessed. Demonstrating superior performance relative to all baselines provides evidence that EM-MNL is the most effective technique for estimating demand under stockout censoring and supports its application and extension in SQ3 and SQ4.

3.4 EM-MNL Estimation Procedure

The EM-MNL algorithm alternates between estimating hidden demand (the E-step) and directly updating preferences (the M-step). Because the M-step uses a direct formula rather than complex inner optimization (Vulcano *et al.* [20]), it runs highly efficiently in $O(TJ)$ time, making it perfectly suited for massive datasets. Full equations are provided in Appendix B.

Three distinct implementation choices are made in this thesis; the full equations they refer to are given in Appendix B. First, the algorithm initializes with $\hat{D}_{jt}^{(0)} = Y_{jt}$ and computes starting preference weights by applying the M-step update (Eq. 9, equivalent to Vulcano Eq. 19) to these values; this provides an empirical "warm start" that accelerates convergence and reduces chance of early local optima. Second, we enforce a tighter convergence criterion, an ℓ_2 norm below $\varepsilon = 10^{-6}$, which guarantees stable stationary points with negligible extra compute cost. Third, to further avoid local optima during the SQ1 sensitivity sweeps (Section 3.5), the model runs five starts per Monte Carlo replication (the warm start plus four seeded random initializations) and retains the highest converged log-likelihood.

3.5 Simulation Study (SQ1): Estimation Accuracy Under Ideal Conditions

This simulation study scales the Monte Carlo validation of Vulcano et al. [20] to roughly one year of daily retail data. Synthetic data are generated using a Poisson-MNL process with $T = 300$ periods and $J = 25$ products. Product availability is Bernoulli ($p_{\text{avail}} = 0.7$), giving a 30% potential stockout rate typical of fashion retail. True preference weights follow a Pareto distribution (shape parameter 1.5, shifted by 1), normalized to $\sum_{j=1}^{J} v_j = 1$ so that demand is skewed and long-tailed, with an implied market share of $s_{\text{true}} = 0.5$. Arrival rates vary by weekday, $\lambda_t = \bar{\lambda}\, w_{t \bmod 7}$, with baseline $\bar{\lambda} = 50$ and multipliers $(1.1, 1.0, 1.0, 1.05, 1.2, 0.8, 0.7)$. Each EM-MNL run uses an observed-sales warm start plus four seeded random initializations and keeps the highest converged log-likelihood (Figure C.1). As a validity check, the setup reproduces the five-product, fifteen-period example of Vulcano et al. [20].

The primary metric is mean absolute percentage bias (MAPB) in total estimated primary demand per product:

$$\text{MAPB}_j = \frac{\left|\hat{N}_j - N_j\right|}{N_j} \times 100\%, \qquad \overline{\text{MAPB}} = \frac{1}{J} \sum_{j=1}^{J} \text{MAPB}_j, \tag{3}$$

where $\hat{N}_j = \sum_t \hat{D}_{jt}$ and $N_j = \sum_t D_{jt}$ are estimated and true total primary demand for product j. MAPB matches the per-product percentage bias reported by Vulcano et al. [20] and its scale-free normalisation permits comparison across the skewed, long-tail demand; its drawback is instability when N_j is small, where a few low-demand products inflate the ratio and can dominate $\overline{\text{MAPB}}$, so the raw-unit MAE is reported alongside as a division-free check. Secondary diagnostics are preference-weight recovery via Pearson correlation, Spearman rank correlation, and NRMSE on normalised weights $p_j = v_j / \sum_k v_k$.

3.5.1 Sub-experiments. Three simulation experiments are conducted: (1) baseline recovery, sweeping horizon length ($T \in \{30, 60, 120, 180, 365, 500, 1000\}$) and availability ($p_{\text{avail}} \in \{0.3, 0.5, 0.7\}$); (2) replication of the validation experiments of Vulcano et al. [20] using Uniform$[0.5, 1]$ preference weights over assortments of 5, 15, and 25 products and horizons of 30, 100, and 300 periods; and (3) sensitivity to market-share misspecification (Appendix C.3). In experiment (1), EM-MNL is compared with the naïve, Poisson GLM, and MLP baselines on identical simulated instances using paired Monte Carlo replications and paired one-sided t-tests on MAPB, with the ML baselines trained under information parity (identical censored inputs and the same censored-cell

imputation task). Machine learning feature construction, hyperparameter grids, convergence verification, and full test statistics are given in Appendix C.1.

3.6 Robustness Analysis (SQ2): Performance Under Assumption Violations

3.6.1 Overview and Shared Design. SQ2 evaluates how EM-MNL accuracy deteriorates as the data progressively violate the estimator's two core assumptions, IIA substitution and Poisson arrivals, both of which fashion retail is known to breach (Section 2). Three misspecification axes are examined, each set by a single severity parameter that scales from a no-violation baseline (the SQ1 Poisson-MNL process) to severe misspecification: Axis 1 (IIA violation) via a Nested Logit DGP with dissimilarity $\lambda_{\text{nest}} \in (0, 1]$; Axis 2 (arrival overdispersion) via a Negative Binomial of dispersion k; and Axis 3 (temporal clustering) via burst-arrival mixtures of intensity α. Each axis reduces to the SQ1 baseline at its no-violation endpoint, and EM-MNL is applied without modification, so any degradation can be attributed to the misspecification. The full DGP formulations and per-axis sweep grids are given in Appendix D.1 (Table D.1).

The shared conditions match SQ1: $T = 300$, $J = 25$, $p_{\text{avail}} = 0.7$, and $s \approx 0.5$ from the same Pareto preference weights (Section 3.5). At each grid point, 200 Monte Carlo replications are run with independent seeds, applying both EM-MNL and the naïve baseline to the misspecified data. The model hyperparameters (initialisation, tolerance, and the five-start protocol of Section 3.4) are the same as in SQ1, so the two studies stay comparable. The primary robustness indicator is the EM-MNL-to-Naïve MAE ratio, computed from the period-level mean absolute error over the full array of per-period, per-product primary demand in raw demand units. A ratio below one means EM-MNL retains its advantage over the naïve baseline; at one or above, its additional complexity is no longer warranted at that misspecification. It is reported alongside the sales-weighted mean absolute percentage error (WMAPE) on per-product seasonal totals (defined in Appendix D.1).

3.6.2 Breakpoint Analysis. For each axis, the breakpoint is defined as the first severity value at which the EM-MNL-to-Naïve MAE ratio reaches 1.0, identified by scanning outward from the no-violation endpoint so that the reported value is the mildest violation at which EM-MNL's relative advantage disappears. This operationalises how far the data can depart from EM-MNL's assumptions before its additional complexity ceases to be warranted. Where the breakpoint is not reached within the swept range, this is noted explicitly; the practical implications are discussed in Section 5.

3.7 Real-World Application (SQ3): Estimation on Multi-Retailer Sales Data

3.7.1 Data and Preprocessing. The WAIR dataset covers daily sales from five Dutch fashion retail chains across 18 store locations over a 93-day window in summer 2024; for confidentiality, chains are labelled A to E and stores numbered within chain (Store D6 is the sixth store of Chain D), and these labels are used throughout. An exploratory analysis of the dataset's sparsity, censoring, and replenishment is given in Appendix A. A product is available in period t when its opening stock exceeds zero ($a_{jt} = \mathbf{1}[\texttt{stock_at_day_start}_{jt} > 0]$), and observed sales are the units sold ($Y_{jt} = \texttt{quantity_sold}_{jt}$); days with no available products are removed, as the EM-MNL likelihood is undefined for empty assortments. Two per-store filters are then applied: phantom SKUs (in stock at least once but never sold) are dropped as they carry no preference signal, and a minimum-sales filter ($\texttt{min_sales} = 3$) restricts the analysis to SKUs with enough in-stock observations to identify a preference weight and test it against the held-out evaluation (Section 3.7.3). The threshold is tested over $\texttt{min_sales} \in \{1, 3, 5\}$ in Appendix E.3.

3.7.2 Applicability Diagnostics. Before estimation, six diagnostic gates calibrated to the operating conditions under which SQ1 and SQ2 demonstrated reliable EM-MNL performance are applied to each store; a store is used for the estimation sample if it passes all six. The gates test overdispersion, period count, product count, non-trivial censoring, assortment variation, and minimum in-stock run length. Their full definitions, thresholds, and the SQ1/SQ2 findings each is calibrated against are given in Appendix E.1.

Of the 18 store chain combinations, 12 pass all six gates and form the estimation sample. Six stores are excluded: Stores A2, D5, and A2 fail because overdispersion exceeds the SQ2 Axis 2 breakpoint, and all three Chain B locations fail the minimum in-stock run length gate (mean run length ≈ 1 day). As a direct check on the gates, the full estimation and held-out pipeline is also run on the six excluded stores (Appendix E.5): the run-length exclusion is confirmed structurally, since no Chain B SKU survives the minimum-sales filter,whereas the overdispersion exclusion is shown to be a conservative precaution, guarding against a failure mode that the held-out test, which without the ground truth cannot detect.

3.7.3 Estimation and Analysis. The EM-MNL algorithm described in Section 3.4 is applied separately to each of the 12 qualifying stores, with all retained SKUs within a store treated as the candidate substitutable assortment. The market share parameter s is fixed at 0.7 as the baseline assumption, corresponding to a 70% chance that an arriving customer makes a purchase when the full assortment is available. To check how sensitive the results are to this assumption, the analysis is repeated for $s \in \{0.3, 0.7, 0.8\}$, and the dependence on s is reported explicitly so that the conclusions rest on patterns that hold across plausible values rather than on a single calibration choice. For each store, the algorithm is initialised using the observed-sales warm start and run until it satisfies the convergence criterion stated in Section 3.4.

Five analyses are conducted on the resulting estimates:

Unmet Demand distribution (per-store). For each SKU j, the uplift ratio $\hat{D}_j^{\text{total}}/Y_j^{\text{total}}$, the ratio of estimated total primary demand to observed total sales, is computed and visualised as a histogram. The store-level summary statistics are the median uplift and the lost-sales fraction $p_{\text{lost}} = (\hat{N}^{\text{total}} - Z^{\text{total}})/\hat{N}^{\text{total}}$, where $\hat{N}^{\text{total}} = \sum_{j,t} \hat{D}_{jt}$ and $Z^{\text{total}} = \sum_{j,t} Y_{jt}$.

EM-MNL vs. Naïve aggregate comparison (per-store). For each store, the aggregate total EM-MNL primary demand is compared to the Naïve estimator (Equation (2)) applied to the same data. This directly addresses whether the additional complexity of the EM-MNL framework produces materially different demand estimates from the simple proportional heuristic.

Temporal patterns. Estimated total product arrivals per day (the product columns of $\hat{D}_{jt}$ summed across SKUs) are plotted against observed total daily sales to verify that the model imputes additional demand on high-stockout days and preserves the weekday rhythm of the underlying arrival process.

Held-out predictive evaluation. Because real sales carry no ground-truth primary demand, recovery bias cannot be measured directly; predictive fit on held-out observed cells is used instead. To assess whether EM-MNL recovers demand signal beyond a descriptive fit, a fraction p of observed in-stock SKU–day cells (j, t) with $a_{jt} = 1$ are masked per store and treated as unobserved. The EM-MNL model is re-estimated on the remaining cells, and imputed values for masked cells are compared against actual observed sales z_{jt}. Evaluation metrics are Mean

Absolute Error (MAE), Root Mean Squared Error (RMSE), mean Poisson deviance, and FracEMbetter (fraction of masked cells where EM-MNL deviance is lower than Naïve). A bootstrap 95% confidence interval on the deviance gap $\Delta_{\text{dev}} = \text{Dev}_{\text{EM}} - \text{Dev}_{\text{Naïve}}$ and a Wilcoxon signed-rank test are used to assess significance. The masking fraction is swept over $p \in \{0.10, 0.15, 0.20\}$ to verify metric stability; baseline results reported at $p = 0.15$.

3.8 Neurosymbolic Extension (SQ4): Rule-Constrained Demand Estimation

While the first three questions (SQ1 to SQ3) rely on the standard EM-MNL estimator, SQ4 explores whether injecting domain knowledge can improve demand estimation, especially when data are sparse. This is addressed using a neurosymbolic approach drawing on Garcez and Lamb [5], treating a symbolic domain rule as a soft constraint so that both the rule and the data jointly guide the estimate.In practice, this rule is applied as a weighted penalty on the loss function, following the semantic-based-regularisation [2, 4, 24]. The rule weight w is kept as a fixed hyperparameter rather than a learned one. A challenge arises because the closed-form EM-MNL updates from Section 3.4 are not differentiable with respect to this kind of penalty, meaning the standard EM-MNL approach cannot easily accommodate gradient-based symbolic regularisation.To resolve this, the estimator is reformulated as a differentiable Poisson-MNL trained via gradient descent. Importantly, when the rule is switched off, this new setup perfectly recovers the original EM-MNL solution. The aim is not to replace the EM-MNL estimator, but to extend it for settings where domain knowledge may provide a useful additional signal. The full objective, identification constraint, and optimiser are detailed in Appendix F.1; this section explains the rule, the estimator, and the evaluation in words.

3.8.1 Symbolic Constraint Derivation. The rule encodes the most basic piece of fashion-assortment structure: a *style* is one garment offered across sizes and colourways, and a customer who cannot find their desired item is far more likely to take another variant of the same style than an unrelated product (Section 2). As a rule, this says that two SKUs in the same style family have similar baseline attractiveness:

$$\forall\, i, j: \quad \text{same_style}(i, j) \;\rightarrow\; v_i \approx v_j. \tag{4}$$

Style groups are taken directly from the WAIR taxonomy (`style_id`), which is rich enough to carry the constraint: 81.8% of style families contain at least two SKUs (Appendix A.3). The rule is deliberately soft, so strong sales evidence can overrule it and w controls how much it is trusted. The rule penalty takes a single input: a list of SKU pairs that should be similar. For each pair, it adds a cost whenever the two SKUs receive different preference weights, pulling their estimates together, with a larger gap incurring a larger cost. Here the list is built from the style taxonomy, pairing any two SKUs in the same style family. The penalty does not depend on where the pairs come from, however, so the list could instead be built from SKUs that are frequently bought together or whose sales move together over time, leaving the objective unchanged. Only the pairs themselves would differ. This thesis uses the style pairing alone.

3.8.2 Differentiable Estimator and the Rule Penalty. The statistical model is identical to EM-MNL's, the customers arrive as a Poisson process and choose under the MNL of Equation (6), so sales of an available product are Poisson. The estimator minimises the negative observed-data log-likelihood plus the rule penalty,

$$\mathcal{L} = \mathcal{L}_{\text{nll}} + w \cdot \mathcal{L}_{\text{rule}}, \tag{5}$$

where $\mathcal{L}_{\text{rule}}$ sums the squared preference-weight gap over every within-style SKU pair and is zero exactly when the rule holds (Appendix F.1). The likelihood does not fix the scale of the preference weights, so the estimator uses same market-share anchor that EM-MNL enforces in its M-step (Section 3.4) into its parameterisation, fixing the scale by construction. The weight $w \geq 0$ has a Bayesian reading: the penalty is the logarithm of a prior that same-style SKUs are similar. At $w = 0$ the penalty vanishes and the estimator minimises the same anchored likelihood EM-MNL addresses, so it must reproduce EM-MNL, the correctness gate for the construction (Section 3.8.3). It is implemented in PyTorch and optimised with Adam followed by an L-BFGS polish.

3.8.3 Evaluation Design. Evaluation is primarily on simulated data, the only setting that supplies the ground-truth primary demand N_j^* against which bias can be measured. The data-generating process places ten products in four style groups whose members share an attractiveness level up to a small spread (about 5%), so the rule genuinely holds, at roughly 50% censoring. Four checks are run, followed by a real-data illustration.

Correctness gate. At $w = 0$ the estimator must reproduce EM-MNL, verified by three hard checks: its primary demand matches an independent maximum-likelihood solver to numerical precision, its likelihood is never worse than EM-MNL's, and its MAPB is within 0.5 percentage points of EM-MNL's (Appendix F.2).

Benefit of a true rule. On the rule-true data, w is swept from 0 to 3000, tracking demand-recovery bias (MAPB against N_j^*, the decisive metric), held-out fit (the SQ3 masked-cell Poisson deviance of Section 3.7.3), and rule satisfaction (the within-style variance of $\hat{v}$, which the penalty drives towards zero). To locate where the benefit stops, the sweep is then extended to $w = 10^5$ and repeated at three within-style spreads $\sigma \in \{0.05, 0.10, 0.20\}$, reading off the interior optimum w^* and the upper weight w_{upper} beyond which even a correct rule degrades recovery (Appendix F.4).

Sample efficiency. To test the prediction that the rule helps most when data are scarce, the demand-recovery comparison is repeated across horizons $T \in \{15, 30, 60, 100\}$, five simulations each, for EM-MNL and the estimator at several rule weights.

Cost of a false rule. The same sweep is run on data with no style structure while the penalty keeps the style labels, so the rule asserts a similarity that does not exist; a constraint that cannot hurt when wrong cannot help when right.

Real-data illustration. The estimator is finally applied to two qualifying SQ3 stores with style groups from the WAIR taxonomy. Real data has no ground-truth demand, so only held-out fit is available and no accuracy claim is made.

3.9 Implementation

All simulation, estimation, and preprocessing routines are implemented in Python. Sales and availability matrices are constructed and manipulated using the Polars data-frame library [18]; all E-step and M-step computations are vectorised over the full $(T \times J)$ matrix using NumPy [8], eliminating per-period Python loops, with statistical tests and held-out evaluation drawing on SciPy [19]. Monte Carlo replications are executed sequentially with deterministic integer seeds to ensure full reproducibility. Visualisations are produced using Matplotlib [9] and Seaborn [22]. The complete implementation, comprising the EM-MNL estimator, the three misspecification-axis simulators, the data preprocessing pipeline, and the evaluation metric functions, is available in the project repository.

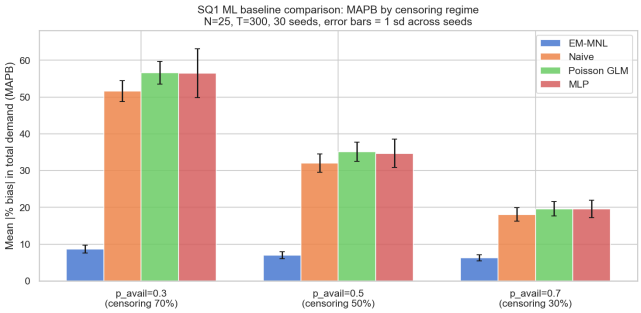


Figure 1: SQ1 ML baseline comparison: per-product demand bias (MAPB) by censoring regime for EM-MNL, the naïve estimator, the Poisson GLM, and the MLP ($J = 25$, $T = 300$, 30 paired Monte Carlo seeds; error bars show one standard deviation across seeds).

4 Results

4.1 SQ1: Baseline Recovery Under Ideal Conditions

Under the Poisson-MNL generative model ($J = 25$ products,{30, 60, 120, 180, 365, 500, 1000}, EM-MNL consistently outperforms the naïve baseline across all 21 (T, availability) combinations ($p < 0.001$, one-sided paired t-test). At the thesis baseline ($T = 300$, a fixed point outside the sweep grid, $p_{\text{avail}} = 0.7$, ≈30% stockout rate), EM-MNL achieves a mean absolute percentage bias (MAPB) of **6.9%** versus **20.7%** for the naïve estimator, a 3× reduction. Period-level MAE is 0.546 (EM-MNL) versus 0.634 (naïve), a 14% improvement concentrated in partial-stockout periods where substitution demand is substantial. The preference-weight recovery confirms near-perfect shape recovery: Pearson $r = 0.998$, Spearman $\rho = 0.957$, NRMSE= 0.054. Figure C.2 shows per-product recovery at the baseline. Across the sweep, MAPB falls as T and availability increase, and EM-MNL stays below the naïve baseline in every condition. Under high censoring ($p_{\text{avail}} = 0.3$, ≈70% stockout rate) at $T = 365$, EM-MNL still reaches 7.9% MAPB, while the naïve estimator's period-level MAE is 1.70× that of EM-MNL.

ML baseline comparison. To test whether a standard supervised-learning approach can recover latent demand without an explicit choice model, two ML baselines (a regularised Poisson GLM and an MLP; Section 3.5.1) are benchmarked against EM-MNL and the naïve estimator on identical simulated instances at $T = 300$, with 30 paired replications per censoring regime. EM-MNL recovers per-product total demand with 6.3%–8.6% MAPB across the three regimes; the naïve estimator records 18.1%–51.7%, the Poisson GLM 19.6%–56.6%, and the MLP 19.6%–56.5% (Figure 1). Every paired comparison favours EM-MNL at $p < 0.001$ on a paired one-sided t-test across the 30 replications (full per-comparison p-values and test protocol in Appendix C.1, Table C.2). Two points from this strengthen the case of EM-MNL. First, both ML models decrease at the rate as the naïve estimator rather than EM-MNL, within a few percentage points of it (and slightly above) at every censoring level. When trained on only the observable censored sales, they inherit the same selection bias, in-stock sales inflated by recaptured substitute demand, and the gap to EM-MNL widens from roughly 3× at 30% censoring to 6.5× at 70%. Second, model capacity does not close the gap: the GLM and the MLP differ by only a few tenths of a percentage point on MAPB, while the MLP is the least stable method across seeds. Additional flexibility cannot recover information that censoring has removed from the data. The structural substitution model is what separates EM-MNL, which deducts recaptured demand from in-stock observations and anchors the demand level through the market-share constraint. Full experimental detail, secondary error metrics, and test statistics are reported in Appendix C.1.

4.2 SQ2: Robustness Under Assumption Violations

Figure 2 summarises the EM-MNL / naïve MAE ratio across three misspecification axes (200 seeds each).

Axis 1: IIA violation (nested-logit DGP). EM-MNL retains its advantage at all tested $\lambda_{\text{nest}} \in \{1.0, 0.8, 0.6, 0.4, 0.2, 0.05\}$; the ratio never reaches 1.0 within the tested range. Absolute error grows as $\lambda_{\text{nest}} \leq 0.2$ and within-nest substitution becomes severe, but EM-MNL remains preferable to the naïve estimator throughout the sweep.

Axis 2: Arrival overdispersion (NegBin DGP). The performance advantage disappears at dispersion parameter $k = 2$; EM-MNL can be trusted in practice for $k \geq 5$. This threshold establishes the Fano-factor eligibility gate applied to the real-data stores in SQ3.

Axis 3: Temporal clustering (burst arrivals). The breakpoint occurs at burst intensity $\alpha = 8$ (5% of periods at 8× normal traffic). The most extreme burst candidate observed in the WAIR data peaks at roughly 2.6× the weekday mean (median 1.7×; Appendix D.2), so even the worst empirical spike sits well below the level at which EM-MNL's advantage is lost.

Empirical motivation from WAIR data. Before testing the system's robustness, real daily arrival data from all 18 WAIR stores was reviewed to ensure the test scenarios were realistic. The findings showed that actual customer arrivals fluctuate much more than the Poisson arrival assumption allows.

The SQ2 sweeps reach well beyond any volatility seen in the WAIR data, so the observed arrivals can be compared against the breakpoints established. Every store sits well inside the burst safe range; only the three most overdispersed stores cross the $k = 2$ threshold and are excluded from SQ3. These exclusions reflect real but extreme variability. Figure D.1 compares the tested range against the observed store values.

4.3 SQ3: Real-World Application

Store eligibility. Of the 18 WAIR store locations spanning five Dutch fashion chains, 12 pass the six SQ2-gates (Figure E.2). Three stores (A2, D5, D6) are excluded by the overdispersion gate, indicating arrival variance exceeding the $k = 2$ breakpoint identified in SQ2. All three Chain B locations fail the minimum in-stock run-length gate (mean run length < 5 days), preventing stable assortment-period identification. The remaining 12 stores span four chains and cover 5,062 SKUs in total after the minimum-sales filter. Running the identical pipeline on the six excluded stores validates the gates from the opposite direction (Appendix E.5): no Chain B SKU survives the minimum-sales filter, so the run-length gate admits no estimable model, while the three overdispersion-excluded stores (A2, D5, D6) still predict held-out sales as well as the qualifying sample. However, this doesn't contradict the Fano gate. The holdout test that fixes the arrival rate is blind to the biased demand recovery during highly variable arrivals. Because of this blind spot, the Fano gate acts as a conservative, theory-driven precaution.

Demand uplift. Table 1 reports chain-level results at the baseline market-share parameter $s = 0.7$. Median per-SKU demand uplift ranges from 1.2×–1.5× (Chain D) to 1.6×–1.8× (Chain A), reflecting the fact that different retail chains experience drastically different levels of hidden demand. Lost-sales fractions (p_{lost}) range from 0.35 to 0.53 across individual stores, with several stores exceeding the 50% mark, so a substantial share of primary demand remains unobserved in censored sales even after restricting to adequately-observed SKUs, at the upper end of the stockout range typical of simulation benchmarks [20]. An example per-SKU uplift distribution (Store D6) and the aggregate EM-MNL versus naïve comparison across stores are shown in Figure E.1.

Sensitivity and cross-chain consistency. Uplift estimates are sensitive to the assumed market-share parameter s: median uplift at $s = 0.3$

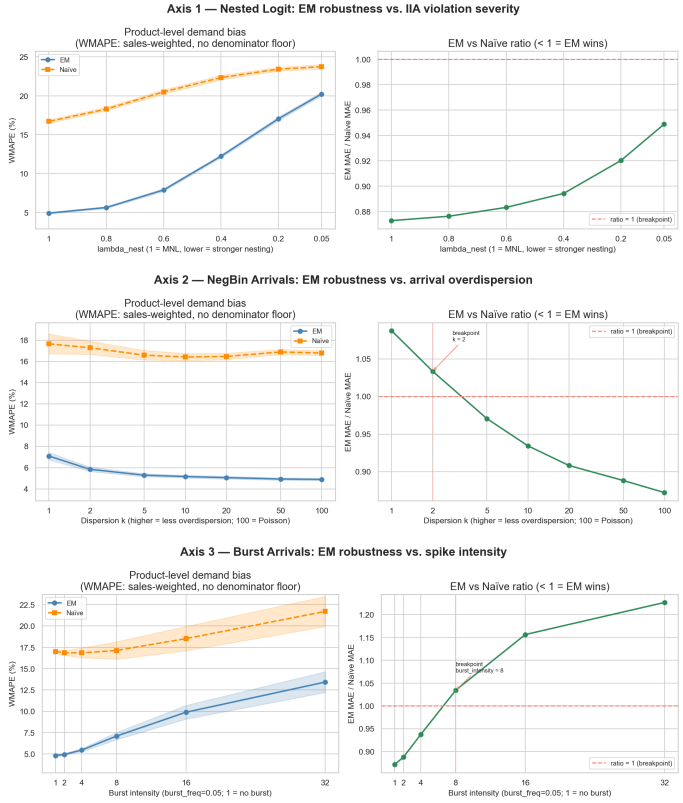


Figure 2: SQ2 robustness sweeps: EM-MNL-to-naïve MAE ratio across each misspecification axis (dashed line marks ratio = 1.0). Top: IIA violation (nested logit, λ_{nest}). Centre: arrival overdispersion (NegBin, k). Bottom: burst intensity (α).

Table 1: SQ3 chain-level summary at $s = 0.7$ (baseline market-share), primary specification `min_sales` = 3. Uplift = $\hat{D}/z$, per-SKU median. p_{lost} = fraction of estimated primary demand representing censored (lost) sales.

Chain	Stores	Median uplift	p_{lost}	SKUs
Chain A	2/3	1.6×–1.8×	0.52–0.53	286
Chain B	0/3	*all excluded (run-length gate)*		
Chain C	3/3	1.4×–1.6×	0.47–0.53	678
Chain D	5/7	1.2×–1.5×	0.35–0.50	3,939
Chain E	2/2	1.3×	0.36	159

is about 3× higher than at $s = 0.8$ across chains. This makes s the largest source of model uncertainty. Within chains, stores that pass the eligibility gates rank SKUs by uplift in broadly the same order (Figure E.3), so demand can be compared across stores for assortment planning. This comparison is meaningful because most style families are carried by more than one store (Appendix A.3). Sensitivity to the minimum-sales filter is reported in Appendix E.3: relaxing the threshold to `min_sales` = 1 readmits weakly identified single-sale SKUs and inflates median uplift to as much as 9.6×, with lost-sales fractions up to 0.94, whereas `min_sales` = 5 leaves as few as 7–11 SKUs in some stores. Setting `min_sales` = 3 avoids both problems, which is why we use it. Appendix C gives the store-level temporal patterns and the full sensitivity sweep.

Held-out predictive evaluation. To test whether EM-MNL recovers demand better than the Vulcano et al. [20] naïve heuristic, a held-out masking evaluation is applied across all 12 qualifying stores. At a masking fraction $p = 0.15$, EM-MNL has lower mean absolute error than the naïve predictor in **all 12 stores** (MAE 0.13–0.17 for EM-MNL versus 0.14–0.18 for the naïve heuristic). For each SKU–day cell, the deviance gap Δ_{dev} is EM-MNL's Poisson deviance minus the heuristic's, so a negative gap means EM-MNL fit that cell more closely. Averaged over each store's cells, this gap favours EM-MNL in 9 of 12 stores, from a slim −0.02 to a large −8.03. The three stores with a positive average are the smallest-assortment ones, where a few cells with near-zero predictions blow up the mean even though most cells still favour EM-MNL (Table E.2). Because that average is so sensitive to those few cells, a bootstrap resamples each store's cells to put a 95% confidence interval around it. These intervals exclude zero for 8 stores, marking a clear result rather than noise, and 7 of the 8 favour EM-MNL. A Wilcoxon signed-rank test compares the per-cell deviances by rank instead of by average, so the extreme cells cannot distort it; it favours EM-MNL at $p < 0.001$ in all 12 stores. The typical held-out cell therefore favours EM-MNL everywhere, even in the stores where a handful of outliers push the average the other way. The MAE advantage and the per-cell win rate hold across masking fractions $p \in \{0.10, 0.15, 0.20\}$, while the mean gap moves around more. Full per-store results are in Appendix E.4.

4.4 SQ4: Neurosymbolic Extension

The differentiable estimator and the style-rule penalty are evaluated on rule-true simulated data, where ground truth is available, and applied to two real stores; the formulation, full tables, and per-figure detail are in Appendix F.

Correctness gate. When the rule weight is zero ($w = 0$), the differentiable estimator behaves exactly like the maximum-likelihood solution. Its per-product demand matches an independent solver to within 5×10^{-7}, far inside the gate threshold of 10^{-3}. Its likelihood is never worse than EM-MNL's, and its bias (MAPB 5.97%) sits within 0.22 percentage points of EM-MNL's (6.19%). The estimator therefore passes the correctness gate.

Setting up this check surfaced a separate finding. EM-MNL's own solution is not quite the constrained maximum-likelihood solution. It is close, but the gap widens as censoring increases, from about 0.7% of demand per product at 10% censoring to 1.9% at 50%. The cause is the ratio-scaling step EM-MNL uses to redistribute demand (Appendix F.2).

Benefit of a true rule. The results show the benefit of using a rule when the rule is actually true. When the rule matches the data, increasing the weight w improves demand recovery. MAPB decreases from 5.97% at $w = 0$ to 3.57% at $w = 3000$, which is better than EM-MNL's 6.19% (Figure F.1). Across this grid, the improvement is one directional. This happens because the test uses a nearly exact rule, with within-style spread $\sigma = 0.05$. In this setting, enforcing equality within a style has almost no cost.

When the same analysis is repeated with larger spreads, using a new dataset each time and comparing against its own EM-MNL baseline, a limit becomes clear (Figure F.2). Once there is real variation between sibling products, the best result occurs at an interior optimum around $w^* \approx 700$. Increasing the weight further eventually becomes harmful, dropping below the EM-MNL baseline, even though the rule itself is still true. This threshold, w_{upper}, decreases as the spread increases, falling from about 10^4 at $\sigma = 0.10$ to around 5×10^3 at $\sigma = 0.20$.

Sample efficiency. The rule helps most when periods are few (Figure F.3). At $T = 15$ it cuts demand-recovery bias from EM-MNL's 14.6% to 9.3%, roughly a third; the gain shrinks as the horizon lengthens and the likelihood carries more on its own. Encoded knowledge substitutes for data, not for the likelihood.

Cost of a false rule. When the style structure is absent but the penalty is applied regardless, bias rises monotonically with w (Figure F.4): from 5.7% at $w = 0$ (matching EM-MNL) to 18.8% at $w = 1000$ and 28.7% at $w = 3000$. This confirms the penalty does genuine work, but the estimator gives no internal warning that a rule is wrong. When applied to the real data a rising held-out deviance is the only external check.

Real-data illustration. On two qualifying stores (Store E2, 159 SKUs; Store D1, 499 SKUs), the $w = 0$ estimator slightly outperforms EM-MNL on held-out deviance, with the naive baseline behind both, and raising w tunes rule adherence without sacrificing fit (Appendix F.7). Real data offers no ground truth, so this is illustration only.

5 Discussion

5.1 Principal Findings

Across all four sub-questions, the answer to the research question is consistent. Within the validated operating range, EM-MNL is an effective and generally robust estimator of primary demand under stockout censoring. It performs better than both the proportional-uplift heuristic and standard supervised baselines. The neurosymbolic extension does not replace it but instead adds structure that can improve stability in some regimes. Its margin of improvement is controlled by how clearly the data allows the true demand parameters to be separated from the censoring process and substitution effects. This includes how much in-stock data is available per product, how dispersed arrivals are, and how correct the substitution model is. When these conditions are strong, EM-MNL is stable and accurate. When they weaken, its gains reduce and sensitivity increases

Five themes emerge from the results in Section 4 that warrant further discussion. Under SQ1, both the regularised Poisson GLM and the MLP closely track the naïve estimator across all levels of censoring. This shows the performance gap to EM-MNL is not driven by model flexibility, but by information structure. Only a model that explicitly represents substitution and enforces a market-share constraint can recover the censored demand signal in a stable way.

Under SQ2, the main failure mode in fashion retail is arrival volatility rather than within-family substitution. EM-MNL can absorb substitution structure within product families, but it treats the arrival process as effectively exogenous and stable. When arrivals become overdispersed, this assumption breaks and identification degrades. In practice, this makes arrival instability the main limiting factor for deploying these models in industry settings, even when substitution behaviour itself is well captured.

Under SQ3, the held-out masking evaluation provides an industry-relevant way to validate demand models without observing true demand directly. This is important in operational settings where ground truth is never available. The results indicate that the recovered uplift reflects genuine signal rather than an artefact of the correction process, supporting its use for decision-making in retail forecasting contexts.

Under SQ4, the neurosymbolic extension introduces differentiable structure without changing the core zero-penalty behaviour of the base estimator. A correct rule effectively substitutes for missing data and improves recovery, particularly in regimes with limited observations per period. A misspecified rule, however, degrades performance in a monotonic way, with no internal diagnostic signal indicating failure.

Constructing the SQ4 correctness gate surfaced a further finding: EM-MNL's fixed point is not the exact scale-constrained maximum-likelihood solution, and the per-product demand gap increases with censoring, from about 0.7% at 10% censoring to about 1.9% at 50% (Table F.1).

This is confirmed this by comparing three different approaches: EM-MNL run to a tolerance of 10^{-12} and converges to a likelihood worse than the gradient solution, while an independent profile-likelihood solver agrees with the differentiable estimator to seven decimal places but not with EM-MNL. The cause is the ratio-scaling E-step, which is exact only in the no-censoring limit. EM-MNL nonetheless remains an excellent approximation, and an analysis against the theory of Abdallah and Vulcano [1] is left to future work.

5.2 Comparison with Prior Work

On simulated data the SQ1 baseline agrees with Vulcano et al. [20] at comparable sample sizes; the contribution is the benchmark itself. Earlier evaluations contrasted EM-MNL with the proportional- uplift heuristic or with one another [3, 20, 21], whereas this study quantifies its gap to the proportional-uplift heuristic the industry uses and to the supervised baselines (Poisson GLM, MLP) a Data scientist may opt for first. That these flexible learners track the naïve estimator rather than EM-MNL has not, to our knowledge, been shown directly, and it justifies preferring a structural choice model under censoring. SQ2 extends evaluation beyond the assumption-satisfying regime to which prior simulations have largely confined themselves [20], showing that within-family substitution is tolerable whereas arrival overdispersion is the genuine constraint on deployment.

On real data, the SQ3 lost-sales fractions sit at the upper end of the 20% to 50% range typical of simulation benchmarks [17, 20]; with the multi-retailer scope of the WAIR data, this distinguishes the study from the predominantly single-retailer prior literature [6, 14, 17], and the masked-cell hold-out answers an evaluation gap noted by the industry partner: how to assess a demand model when true demand is unobservable. The SQ4 extension situates EM-MNL within the neurosymbolic programme [5], demonstrating the trade-off between built-in assumptions and misspecification risk in an operations-research estimator rather than the deep classifiers where it is usually studied. It shares motivation with optimisation-based estimators, most directly the Subramanian and Harsha MIP [15], but differs in being a differentiable reformulation of EM-MNL itself: it reproduces standard recovery at zero penalty weight (the correctness gate, Appendix F.2) and imposes a soft penalty over the style-family substitution graph rather than hard mixed-integer constraints.

5.3 Limitations

The real-data validation establishes predictive fit, not recovery. The masked-cell hold-out of Section 4.3 confirms that EM-MNL predicts observed in-stock sales more closely than the naïve heuristic and that qualifying stores rank SKUs by uplift in broadly the same order, but it cannot verify the absolute primary demand the method recovers, and because it holds the arrival rate fixed it is does not know the overdispersion-induced recovery bias the Fano gate guards against (Appendix E.5). Offline sales supply no ground-truth demand against which that bias could be checked; the online-retail direction of Section 5.4 is the natural remedy. The empirical scope is also narrow. The data come from one aggregator (WAIR) over a single 93-day window in summer 2024, so the study is multi-retailer but not multi-season, multi-region, or multi-format. The work discusses a permanent-stockout regime: in 87.4% of stocked location–SKU pairs the product is received once and never replenished (Appendix A.2, Figure A.4), a pattern that need not transfer to transient-stockout or online settings. The minimum-sales filter compounds this: with so little in-stock data, the sparsest, most-censored SKUs leave too few observations to identify a preference weight and are forced to be excluded, even though they are exactly the items the method exists to recover (Appendix E.3).

The symbolic extension uses a single style-coherence rule whose weight is set by hand rather than learned, and the estimator returns no internal diagnostic when that rule is wrong, so its correctness remains

the practitioner's responsibility (Section 4.4, Appendix F.6); only style coherence was tested. This sits within a broader structural simplification: the SQ3 estimation of Section 3.7.3 pools every retained SKU in a store into one MNL choice set, applying the IIA assumption across unrelated categories, and SQ4 addresses this only partially by injecting within-style structure. The learned co-occurrence substitution graph outlined in Section 5.4 would address it more completely. Underlying all of these, the dependence on an assumed market-share parameter is the dominant source of uncertainty in the absolute uplift levels (Sections 4.3, 5.1, and 5.4), and the SQ2 gates are calibrated on isolated misspecification axes, so compound violations remain untested.

5.4 Future Work

There are several directions for future development. The main source of SQ3 uncertainty could be removed by fixing the market-share parameter s from an outside footfall or conversion signal. The SQ4 finding that EM-MNL's fixed point only approximates the constrained MLE is a reason to make the differentiable estimator the main solver, because the gap is widest under the heavy censoring that fashion retail tends to see. The single style-coherence rule could be widened into a co-occurrence or correlation graph that gives soft, editable substitution groups in place of rigid style and size categories; the hard part is learning or checking such structure rather than assuming it, because a wrong rule detriments performance. Finally, a controlled restocking trial in WAIR's own stores, where selected SKUs are deliberately kept in stock and the resulting sales observed, would give a near-ground-truth check on the uplift magnitudes the current observational data cannot supply.

6 Conclusion

When products sell out, observed sales understate true demand: the stocked-out item records no further sales, while its would-be buyers either substitute or leave. This thesis asked how well EM-MNL, and a simple neurosymbolic extension of it, recover primary demand from censored fashion-retail sales, and which gives greater accuracy, robustness, and interpretability.

On simulated data with known ground truth, EM-MNL recovers per-product demand to within a few percent and beats a proportional-uplift heuristic, a regularised regression, and a neural network, all of which inherit the censoring bias (SQ1). It absorbs substitution within style families but loses its edge once arrivals grow overdispersed, so arrival volatility, not substitution structure, is the real limit on fashion deployment (SQ2). Across twelve qualifying WAIR stores it recovers a large share of hidden demand, and a held-out test needing no ground truth confirms it predicts better than the heuristic in every store (SQ3). The extension keeps EM-MNL as its core and adds a style-similarity rule as a soft penalty: it improves recovery when the rule holds, most when data are scarce, is untouched when switched off, and harms recovery when the rule is false (SQ4). Building its correctness check also showed that EM-MNL's fixed point is not the exact scale-constrained maximum-likelihood solution, the gap widening under heavier censoring.

The work gives practitioners a first basis for judging whether EM-MNL suits a given fashion dataset. Its limits are a single aggregator over one summer window and an absolute uplift that depends on an assumed market share, so the recovered figures show the size of the correction under that assumption, not a fixed total. The clearest next steps are to fix market share from an external footfall signal, to make the differentiable estimator the main solver given the censoring-dependent drift, and to run a controlled restocking trial for a near-ground-truth check.

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Appendix A Exploratory Data Analysis

An exploratory analysis of the full WAIR dataset (five retail chains, 18 store locations, 35,231 unique SKUs over the 93-day Q3 2024 window) was done before the estimation work began. This appendix is not a full record of the analysis. It reports the subset of findings that directly motivate choices in the main text, so that design decisions can be traced back to an observed property of the data.

A.1 Store Heterogeneity and Sales Sparsity

Figure A.1 summarises the dataset at store level. Total sales volume and assortment size vary greatly across the 18 stores, and the proportion of SKU–day rows recording a non-zero sale is below 2% in every store. This heterogeneity is why estimation is done independently per store and why per-store applicability gates are required (Section 3.7.2): a pooled specification would be dominated by a handful of high-volume stores while masking the lack of data in the smaller ones.

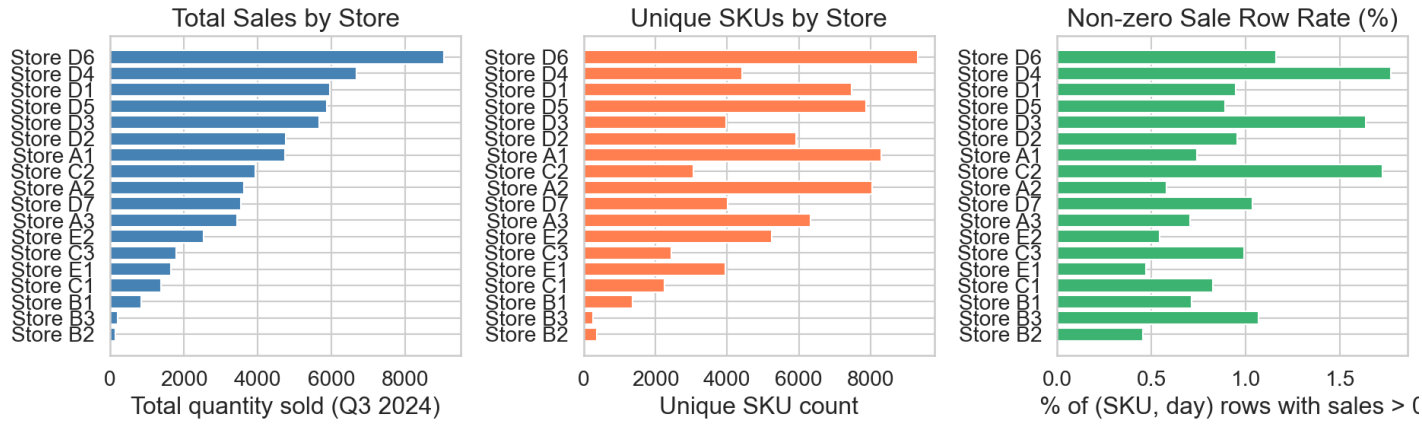


Figure A.1: Store-level overview of the WAIR dataset: total units sold (left), unique SKUs carried (centre), and the percentage of SKU–day rows with a non-zero sale (right), per store.

Figure A.2 shows the SKU-level data. After removing tenant-dead SKUs, the median SKU sells exactly 1 unit over the full quarter, 21.8% of stocked SKUs record zero sales, and 88.6% sell five units or fewer. This extreme sparsity is the binding constraint on per-SKU demand estimation in SQ3: it motivates the phantom-SKU exclusion and the minimum-sales filter of Section 3.7.1. Since the method exists to recover demand for exactly these low-sales, censored items, the threshold that excludes them is varied and tested rather than fixed by assumption.

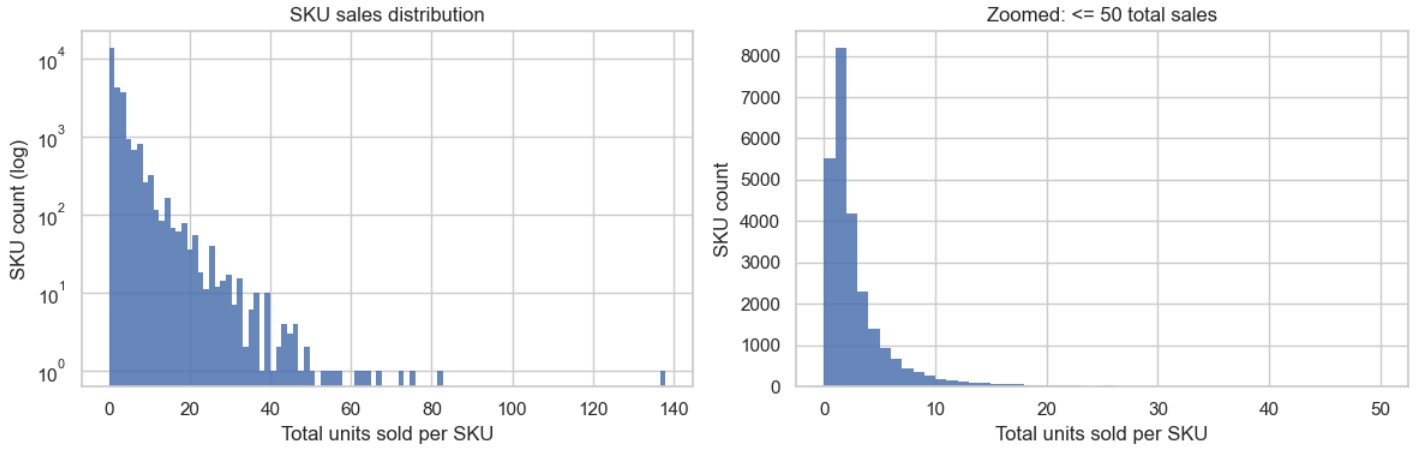


Figure A.2: Distribution of total Q3 2024 sales per SKU (tenant-dead SKUs excluded; n = 25,379). Left: full distribution. Right: zoomed to SKUs with at most 50 total sales.

A.2 Stockout Censoring and Replenishment

Figure A.3 reports the stockout rate per store, the share of SKU–day cells in which a listed product was unavailable. Censoring happens in every store and varies widely across stores, which establishes the premise of the thesis: observed sales systematically understate demand, and the size of the correction needed differs by store.

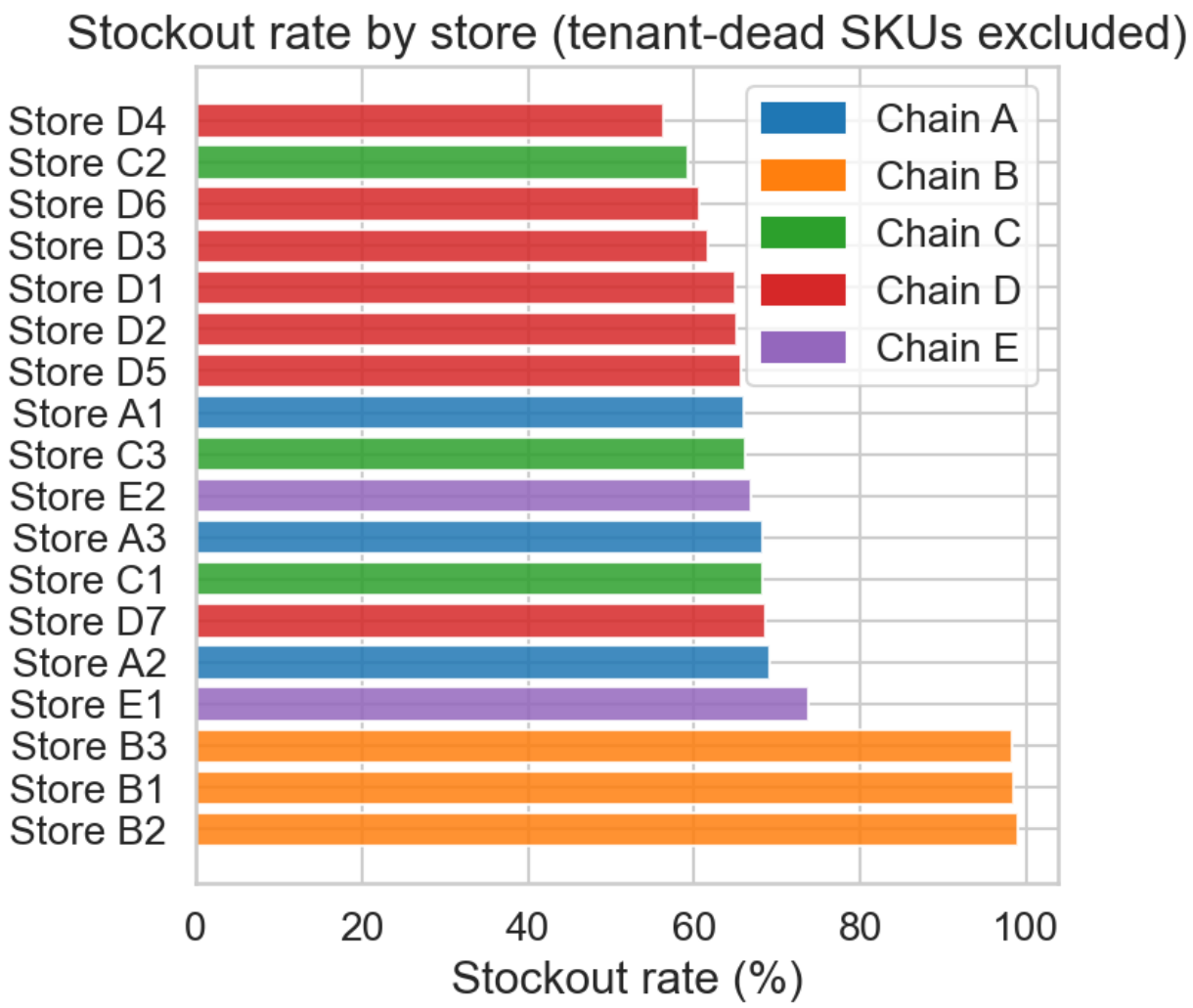


Figure A.3: Stockout rate by store (tenant-dead SKUs excluded): the fraction of SKU–day cells in which a listed SKU had no opening stock.

Figure A.4 shows that this censoring is structural. In 87.4% of stocked (location, SKU) pairs the product received stock exactly once and was never replenished within the window. A stockout is therefore usually permanent, so demand for a stocked-out product must be inferred from its earlier in-stock periods and from substitution to available alternatives. This is the setting EM-MNL is designed for, but it works only when a product is in stock long enough beforehand to reveal its demand. When a product stocks out almost immediately after arriving, no usable in-stock window remains, which is what the minimum in-stock run length gate of Section 3.7.2 screens out.

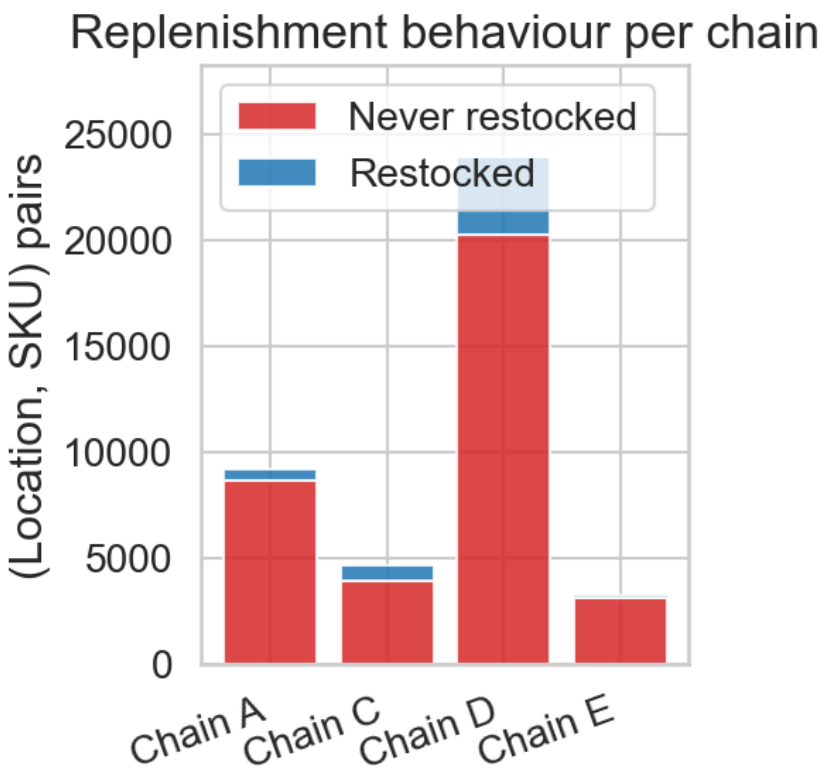


Figure A.4: Replenishment behaviour per retail chain: stocked (location, SKU) pairs split into those never restocked (received stock exactly once) versus those restocked at least once within the window.

A.3 Product Hierarchy and Style Structure

The SQ4 extension reallocates demand within substitution groups in product taxonomy (Section 3.8). Figure A.5 shows that the taxonomy can support such groups. Of the 8,305 distinct style families, 81.8% contain at least two SKUs, so within-style substitution is defined for most of the assortment.

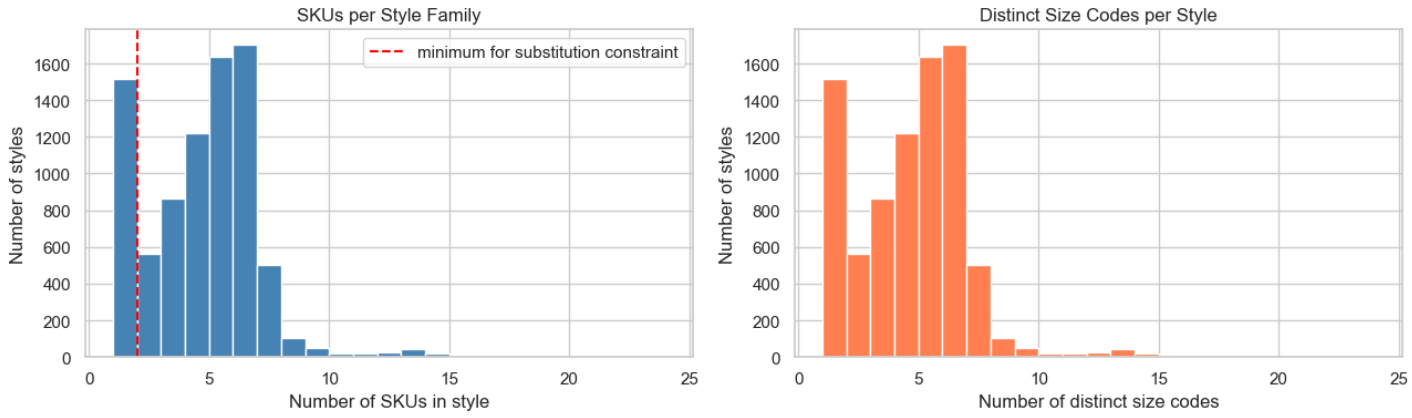


Figure A.5: Product hierarchy structure: SKUs per style family (left) and distinct size codes per style (right), over 8,305 unique styles.

Figure A.6 shows how broadly styles are shared across stores. Assortments are largely local: 2,059 of the 8,305 styles are carried by a single store and none is carried by all 18. The majority of styles nevertheless appear in two or more stores, which is what makes the cross-store consistency analysis of the SQ3 results meaningful.



Figure A.6: Number of stores carrying each style family (8,305 styles across 18 stores).

Appendix B EM-MNL Algorithm Details

This appendix provides the full choice probabilities, E-step and M-step equations, the outside-good imputation, and the iterative structure of the EM-MNL estimator summarised in Section 3.4. The estimator is the model of Vulcano et al. [20]; the implementation choices specific to this thesis (initialisation, convergence tolerance, multi-start protocol) are stated in Section 3.4. To keep things simple, let $V = \sum_{j=1}^{J} v_j$ be the total product weight under full assortment and $V_t^{\text{avail}} = \sum_{k \in S_t} v_k$ the total weight of available products in period t.

Arrivals in each period t follow a Poisson distribution with rate λ_t, and each arriving customer selects from the available assortment S_t according to MNL probabilities set by the preference weight vector $\mathbf{v}$. The probability that a customer chooses product j when assortment S_t is offered is

$$P_j(S_t, \mathbf{v}) = \frac{v_j}{\sum_{k \in S_t} v_k + 1}, \tag{6}$$

where the denominator includes a normalised no-purchase option with weight $v_0 = 1$.

B.1 E-step

Given the current preference weights $\mathbf{v}^{(k)}$, the latent primary demand $\hat{D}_{jt}^{(k)}$ is imputed separately for available and unavailable products. For available products ($a_{jt} = 1$), observed sales are scaled down to deduct demand recaptured from stocked-out alternatives:

$$\hat{D}_{jt}^{(k)} = Y_{jt} \cdot \frac{V_t^{\text{avail}} + 1}{V + 1}. \tag{7}$$

For unavailable products ($a_{jt} = 0$), latent primary demand is imputed from the period's total observed sales, weighted by the product's preference share under the full assortment:

$$\hat{D}_{jt}^{(k)} = \frac{v_j}{V + 1} \cdot \frac{V_t^{\text{avail}} + 1}{V_t^{\text{avail}}} \cdot \sum_{i \in S_t} Y_{it}. \tag{8}$$

Equations (7) and (8) correspond to Equations (7) and (6) respectively in [20].

B.2 M-step

Given the imputed product totals $N_j^{(k)} = \sum_t \hat{D}_{jt}^{(k)}$, preference weights are updated in closed form:

$$v_j^{(k+1)} = \frac{N_j^{(k)}}{r \sum_{i=1}^{J} N_i^{(k)}}, \qquad j = 1, \ldots, J. \tag{9}$$

This is equivalent to [20] Eq. (14) under the market-share constraint $N_0 = r \cdot \sum_{i=1}^{J} N_i$. The implementation uses $r \cdot \sum_{i=1}^{J} N_i^{(k)}$ in the denominator rather than the imputed $N_0^{(k)}$ from the E-step; the two are identical when the market-share constraint holds, and using the constrained form enforces the constraint at every iteration regardless of any imputation error in the outside-good total.

B.3 Outside-good imputation

The imputed no-purchase total in period t is computed as

$$\hat{D}_{0t}^{(k)} = \frac{\sum_{j=1}^{J} \hat{D}_{jt}^{(k)}}{V}, \tag{10}$$

that is, the period's total imputed product demand divided by V rather than by $V + 1$. This reflects the outside-good ratio $r = 1/V$ at full assortment (with $v_0 = 1$) and matches [20] Eq. (10). The quantity $\hat{D}_{0t}^{(k)}$ contributes to the imputed period arrival rates $\hat{\lambda}_t = \sum_{j=0}^{J} \hat{D}_{jt}^{(k)}$ but does not enter the M-step update in Equation (9). Convergence of the preference weights is therefore independent of this imputation; the imputation matters only for downstream interpretation of $\hat{\lambda}_t$ and the no-purchase total (for example, in the market-share sensitivity analysis of Appendix C.3).

B.4 Iterative structure

The algorithm is initialised with $\hat{D}_{jt}^{(0)} = Y_{jt}$ and initial preference weights computed by applying Equation (9) to these starting values ([20] Eq. 19). Iterations alternate between the E-step (Equations 7–8) and the M-step (Equation 9). The loop terminates when the ℓ_2 norm of successive preference-weight updates falls below the tolerance $\varepsilon = 10^{-6}$:

$$\left\| \mathbf{v}^{(k)} - \mathbf{v}^{(k-1)} \right\|_2 < 10^{-6}. \tag{11}$$

Figure B.1 illustrates this structure.

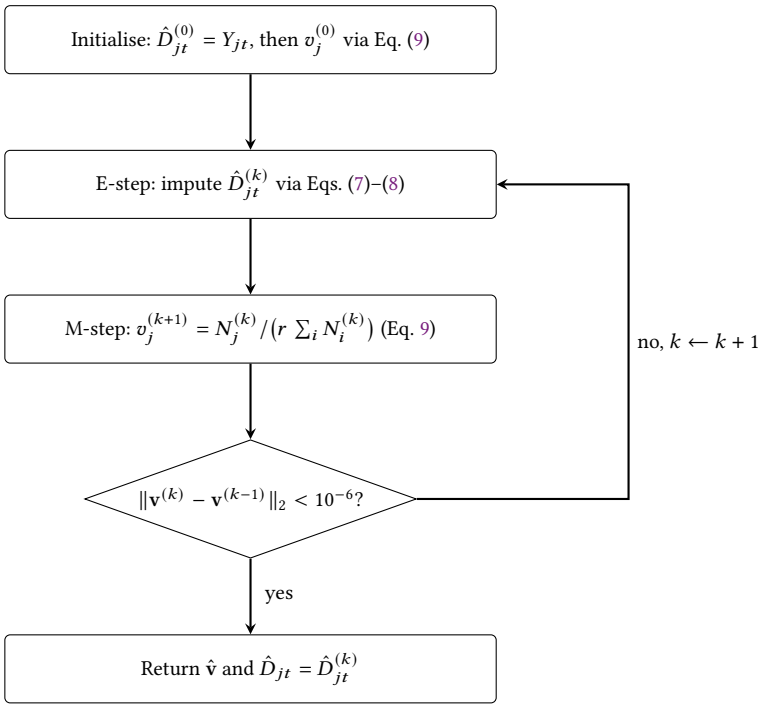


Figure B.1: Iterative structure of the EM-MNL algorithm. The loop alternates E-step and M-step until the ℓ_2 norm of successive preference-weight estimates falls below 10^{-6}.

Appendix C SQ1: Simulation Study

C.1 ML Baseline Comparison: Experimental Detail

This subsection records the experimental detail behind the ML baseline comparison reported in Section 4.1. Both ML models are fitted to in-stock product–period cells only; out-of-stock cells are excluded from training because their observed sales are zero by construction rather than by lack of demand.

Features. Each cell (j, t) is described using product indicator variables, the number of products in stock during period t, a leave-one-out traffic measure (total sales in the period minus the sales of that cell, so the cell's own outcome is not used as a feature), and a time index. A product availability flag is also included for compatibility, although it is constant in the in-stock training data. One source of bias remains. During periods with many stockouts, observed traffic is lower because some customers choose not to buy anything when their preferred product is unavailable. As a result, the features are affected by the same censored sales process seen in observational data. This is part of the comparison rather than a flaw in the method.

Hyperparameter search. The Poisson GLM is tuned using grid search over regularisation values of $\alpha \in \text{logspace}(-3, 2, 8)$ with five-fold cross-validation. Performance is measured using mean Poisson deviance. The MLP uses two hidden layers with 64 units each and is tuned over $\alpha \in \text{logspace}(-4, 0, 5)$ and initial learning rates of $10^{-3}, 10^{-2}$. Three-fold cross-validation is used to reduce computation time. Mean squared error is used as the scoring metric because Poisson deviance cannot be calculated when the network produces negative values. Final MLP predictions are clipped at zero. Feature standardisation is carried out within each cross-validation pipeline, meaning that scaling values are estimated using only the training data. Demand is then estimated using the na"ive imputation approach: observed sales are used when a product is in stock, and the clipped model prediction is used when a product is out of stock.

EM-MNL protocol and convergence verification. EM-MNL is run using the observed sales as a warm start together with four additional random starting points. The maximum number of iterations is set to 500, with a convergence tolerance of 10^{-6}. The solution with the highest converged log-likelihood is selected. Across all 90 instances (3 regimes × 30 seeds), every run converged before reaching the iteration limit. Figure C.3 shows the ℓ_2 update norm decreasing until it reaches the convergence threshold.

Results. Table C.1 presents the full comparison. One important point should be noted. Under the lowest level of censoring, the naive estimator performs almost the same as EM-MNL when predicting stocked-out cells, with per-cell MAE values of 0.189 and 0.187, respectively. This means that EM-MNL does not gain its advantage through better per-cell imputations in this setting. Instead, its benefit appears in the product-level totals, where the na"ive method systematically overestimates demand by counting in-stock sales too highly. The ML baselines perform worse than the na"ive estimator on stocked-out cells in all censoring regimes.

Table C.1: SQ1 ML baseline comparison at $T = 300$, $J = 25$, 30 paired Monte Carlo replications per censoring regime. MAPB on per-product total demand (mean and standard deviation across replications, and ratio to EM-MNL); period-level MAE and RMSE over all cells; MAE on stocked-out cells only (MAE OOS).

Censoring	Method	MAPB (sd)	vs EM-MNL	MAE	RMSE	MAE OOS
70%	EM-MNL	8.6% (1.07)	1.00×	0.370	0.594	0.270
	Naïve	51.7% (2.83)	5.99×	0.626	0.927	0.483
	Poisson GLM	56.6% (3.09)	6.56×	0.666	0.966	0.540
	MLP	56.5% (6.61)	6.55×	0.667	0.970	0.541
50%	EM-MNL	7.0% (0.98)	1.00×	0.433	0.667	0.214
	Naïve	32.1% (2.48)	4.58×	0.584	0.910	0.302
	Poisson GLM	35.1% (2.62)	5.03×	0.611	0.927	0.355
	MLP	34.7% (3.84)	4.96×	0.609	0.925	0.351
30%	EM-MNL	6.3% (0.80)	1.00×	0.541	0.789	0.187
	Naïve	18.1% (1.79)	2.87×	0.621	0.933	0.189
	Poisson GLM	19.6% (1.96)	3.11×	0.631	0.937	0.225
	MLP	19.6% (2.40)	3.11×	0.636	0.938	0.241

Significance testing. Each EM-MNL method was compared with the other estimators using a paired one-sided t-test on replication-level MAPB (`scipy.stats.ttest_rel`). A paired test is appropriate because all four methods were evaluated on the same simulated datasets across the 30 seeds, meaning that only the within-seed differences contributed to the statistical test. The null hypothesis stated that the methods had equal mean MAPB, while the alternative hypothesis was that EM-MNL had a lower mean MAPB ($n = 30$ replications, df $= 29$). Table C.2 reports the resulting p-values for all nine comparisons. In every case, the null hypothesis was rejected at $p < 0.001$ (all values were below 10^{-22}). These very small p-values reflect the large and consistent MAPB differences across seeds rather than a large sample size. The exact values are reported to document the statistical test and should not be interpreted as measures of effect size.

C.2 Demand Recovery and Convergence

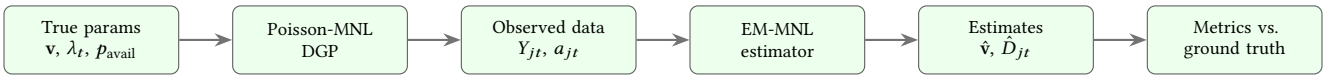


Figure C.1: Simulation pipeline for evaluating EM-MNL demand recovery.

Table C.2: SQ1 ML baseline comparison: paired one-sided t-test p-values on replication-level MAPB. The alternative hypothesis is that EM-MNL attains lower mean MAPB than the competitor ($n = 30$ paired replications per regime, $\mathrm{df} = 29$).

Censoring	EM-MNL vs Naïve	EM-MNL vs Poisson GLM	EM-MNL vs MLP
70%	5.2×10^{-36}	4.0×10^{-36}	9.8×10^{-27}
50%	4.4×10^{-31}	7.7×10^{-32}	1.9×10^{-27}
30%	1.5×10^{-24}	8.9×10^{-25}	8.2×10^{-23}

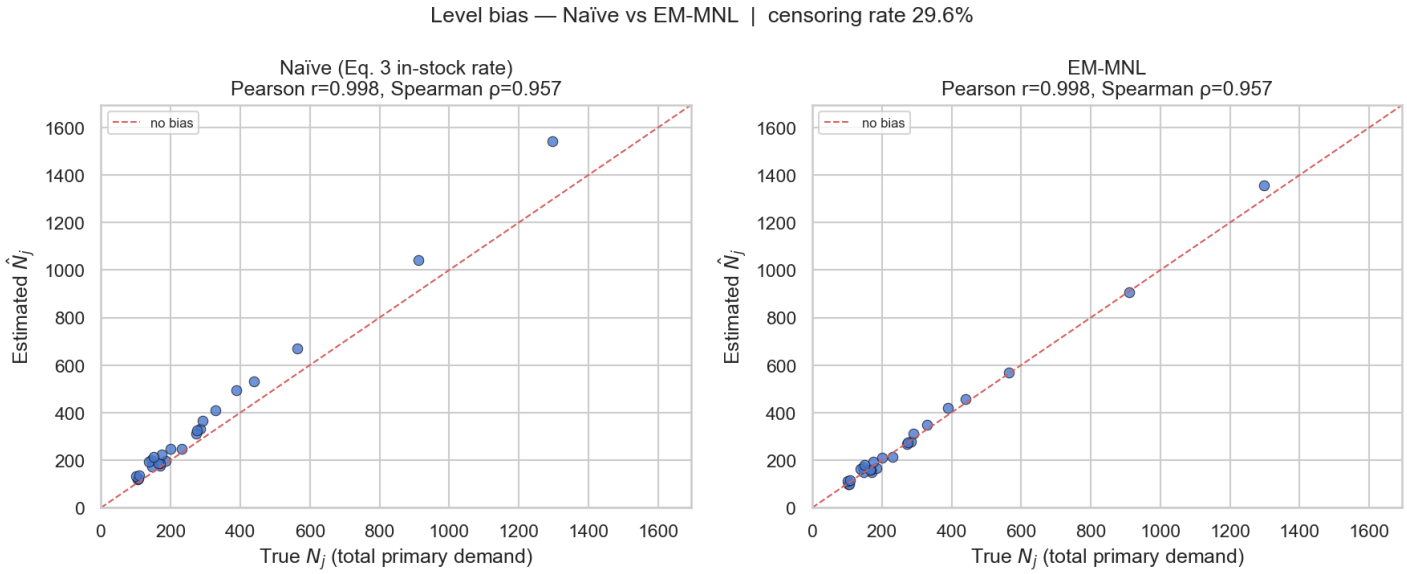


Figure C.2: Estimated vs. true primary demand N_j per product for the naïve and EM-MNL estimators (baseline: $T = 300$, $p_{\text{avail}} = 0.7$, $J = 25$).

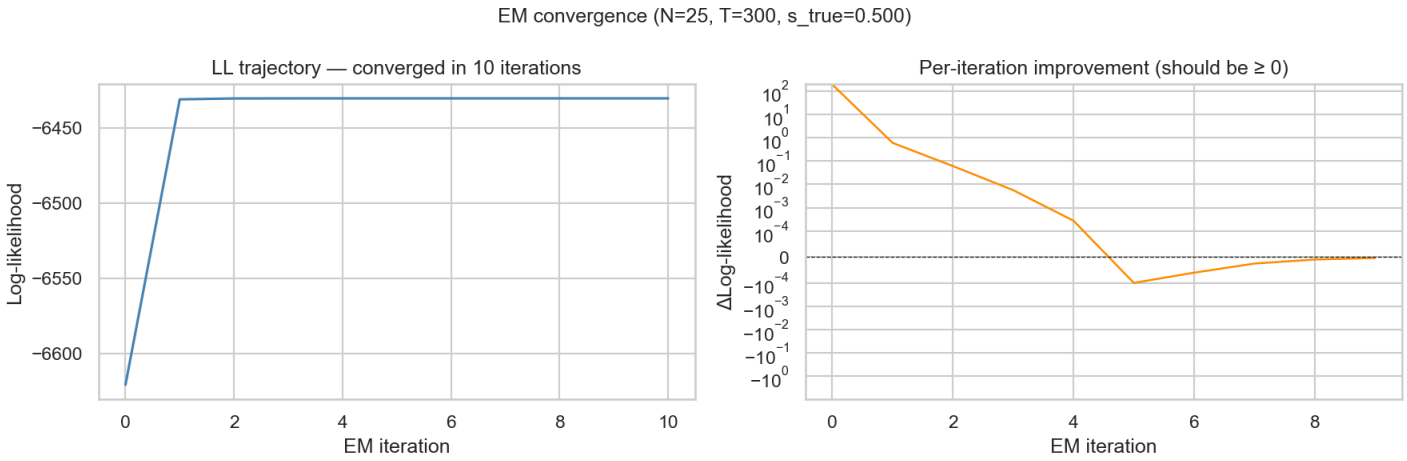


Figure C.3: EM-MNL convergence behaviour across Monte Carlo seeds: ℓ_2 norm of successive preference-weight updates versus iteration count.

C.3 Market-Share Sensitivity

The SQ1 notebook also repeats the market-share misspecification check of Vulcano et al. [20]. The experiment uses $J = 10$, $T = 500$, $\lambda = 50$, $p_{\text{avail}} = 0.7$, and preference weights drawn from Uniform$[0.5, 1]$. The true market share implied by the sampled preferences is $s_{\text{true}} = 0.876$, and the estimator is rerun at $\hat{s} \in \{0.8s_{\text{true}}, 0.9s_{\text{true}}, s_{\text{true}}, 1.1s_{\text{true}}, 1.2s_{\text{true}}\}$ across 100 Monte Carlo seeds. As expected, misspecifying s changes the recovered preference scale and the implied arrival rate, while product-level primary-demand bias remains comparatively stable: the maximum mean absolute primary-demand bias across the five misspecification levels is 8.11% (Figure C.4). This check is treated as a supplementary calibration validation rather than as one of the headline SQ1 estimator comparisons.

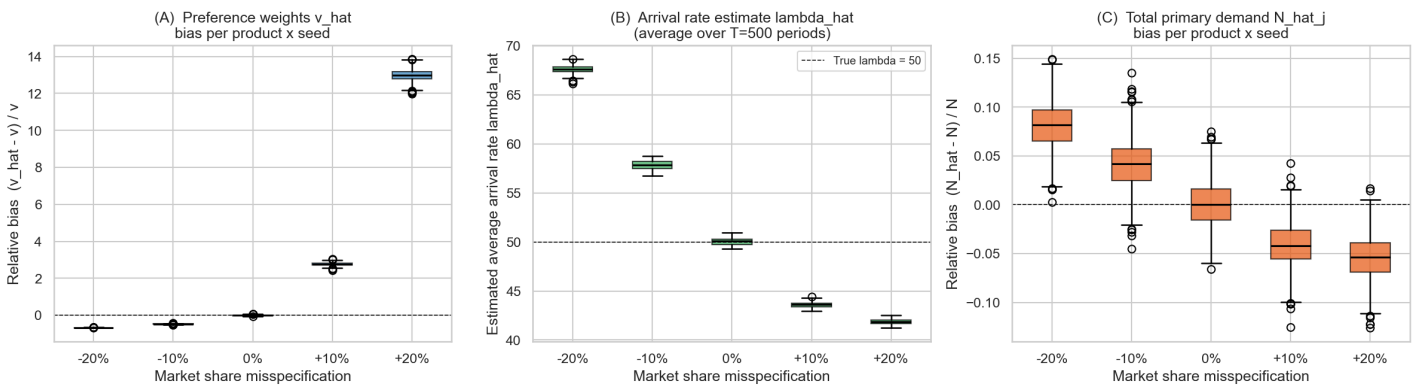


Figure C.4: SQ1 market-share misspecification check (reproduction of the Vulcano et al. [20] sensitivity exercise): recovered preference weights, arrival rate, and total primary demand as the supplied market-share anchor $\hat{s}$ is varied.

Table D.1: Summary of the three misspecification axes for the robustness study (SQ2). Each axis reduces to the SQ1 Poisson-MNL baseline at its no-violation endpoint.

Axis	DGP	Severity parameter	Sweep values
IIA violation	Nested Logit	$\lambda_{\text{nest}} \in (0, 1]$ ($\lambda_{\text{nest}} = 1 \Rightarrow$ standard MNL)	{1.0, 0.8, 0.6, 0.4, 0.2, 0.05}
Arrival overdispersion	Negative Binomial	$k \in \{1, 2, \ldots\} \cup \{\infty\}$ ($k \to \infty \Rightarrow$ Poisson)	{∞, 50, 20, 10, 5, 2, 1}
Temporal clustering	Burst arrivals	$\alpha \in [1, 32]$, $f = 0.05$ fixed ($\alpha = 1 \Rightarrow$ Poisson)	{1, 2, 4, 8, 16, 32}

Appendix D SQ2: Robustness Study

D.1 Misspecification DGP Formulations

This subsection explains the full definitions of the three misspecified data generating processes introduced in Section 3.6 and summarised in Table D.1. Each axis becomes the SQ1 Poisson-MNL baseline when there is no violation. EM-MNL is used without any changes, so any drop in performance can only be caused by the misspecification. For each axis, estimator bias is measured against the expected full-assortment demand produced by the misspecified DGP itself, rather than against an assumed MNL truth. This means that any loss in performance shows a failure to recover the DGP's own demand, not a failure to reproduce MNL.

The product-level seasonal demand bias reported alongside the EM-MNL-to-Naïve MAE ratio is summarised using the sales-weighted mean absolute percentage error, WMAPE $= \left(\sum_j |\hat{N}j - N_j| \Big/ \sum_j N_j\right) \times 100\%$, where $\hat{N}j = \sum_t \hat{D}jt$ and $N_j = \sum_t Djt$. This combines demand across all periods by measuring each product using its seasonal total and then scaling by the total realised demand.

Because WMAPE combines all periods into a single seasonal total, it rewards an estimator that recovers the correct total demand for each product, even if that demand is assigned to the wrong periods. In contrast, the period-level MAE used in the EM-MNL-to-Naïve ratio directly penalises this type of timing error. The naïve estimator is also almost exact for in-stock observations by construction. As a result, the two metrics do not always change in the same way and are reported together rather than used interchangeably. SQ1 reports the unweighted MAPB, while SQ2 reports the sales-weighted WMAPE. To allow comparison, both measures are shown at the shared no-violation anchor, the SQ1 Poisson-MNL baseline with $N = 25$, $T = 300$, and $p_{\text{avail}} = 0.7$, which is the point to which every SQ2 axis reduces when there is no violation. At this anchor, EM-MNL achieves a MAPB of 6.9% and a WMAPE of 4.9%, compared with 20.7% and 16.8% for the naïve estimator. WMAPE is lower in both cases because sales weighting reduces the influence of low-demand products, which tend to have the largest percentage errors. However, both measures lead to the same conclusion that EM-MNL performs better than the naïve estimator, and the corresponding EM-MNL-to-naïve MAE ratio at this anchor is approximately 0.87.

Axis 1: IIA violation via Nested Logit. Products are grouped into five nests of five, representing style families. Within-nest correlation is governed by the dissimilarity parameter $\lambda_{\text{nest}} \in (0, 1]$. When $\lambda_{\text{nest}} = 1$, the model reduces to standard MNL. As λ_{nest} decreases toward 0, substitution becomes increasingly concentrated within nests, strengthening the IIA violation. The number of nests is fixed to five to separate correlation strength from partition size. Choice probabilities are computed in log-space for numerical stability.

Axis 2: arrival overdispersion via Negative Binomial. The Poisson arrival process is replaced with a Negative Binomial:

$$A_t \sim \text{NegBin}(\lambda_t, k), \qquad \text{Var}(A_t) = \lambda_t + \frac{\lambda_t^2}{k} \tag{12}$$

where $k > 0$ controls overdispersion. As $k \to \infty$, the process converges to Poisson. Smaller values of k increase variance across periods without introducing temporal dependence (handled separately in Axis 3). Since the EM E-step does not explicitly use the arrival distribution, the impact of overdispersion arises only through deviations from the Poisson mean–variance assumption over time.

Axis 3: temporal clustering via Burst Arrivals. In 5

$$A_t \sim \text{Poisson}\big(\lambda_t \cdot (1 + B_t(\alpha - 1))\big). \tag{13}$$

The burst intensity α is varied over 1, 2, 4, 8, 16, 32, ranging from no effect ($\alpha = 1$) to extreme spikes (32× baseline). The key violation comes from the mixture structure: each period's effective arrival rate is not deterministic but is instead randomly drawn from a two-state process (baseline vs. burst). This induces a stochastic regime-switching data-generating process, whereas the EM-MNL estimator assumes a single smooth and deterministic evolution of λ_t. As a result, the estimator cannot represent abrupt jumps in intensity and systematically misattributes burst-induced variation to gradual changes in the underlying rate.

D.2 Empirical Arrival Distribution Analysis

EDA was done to explore the arrival distributions across WAIR stores to verify that the SQ2 robustness tests cover realistic violation ranges. Figure D.1 addresses overdispersion (Axis 2): it contrasts the Poisson assumption, the empirical within-weekday distribution for Store D6 (Fano = 16.51, MOM $\hat{k} = 6.3$), and the NegBin(λ, k) family. The most overdispersed store sits above the $k = 2$ breakpoint and just above the $k \geq 5$ practical safe zone established in Section 4.2, so the empirically observed range is covered by the SQ2 sweep. Figure D.2 addresses temporal clustering (Axis 3): it contrasts a Poisson time series (no bursts by construction), the WAIR empirical time series with burst candidates identified (10.8% of days above 1.5× the weekday mean, median intensity 1.7×, maximum 2.6×), and a simulated burst DGP at the SQ2 breakpoint ($\alpha = 8$).

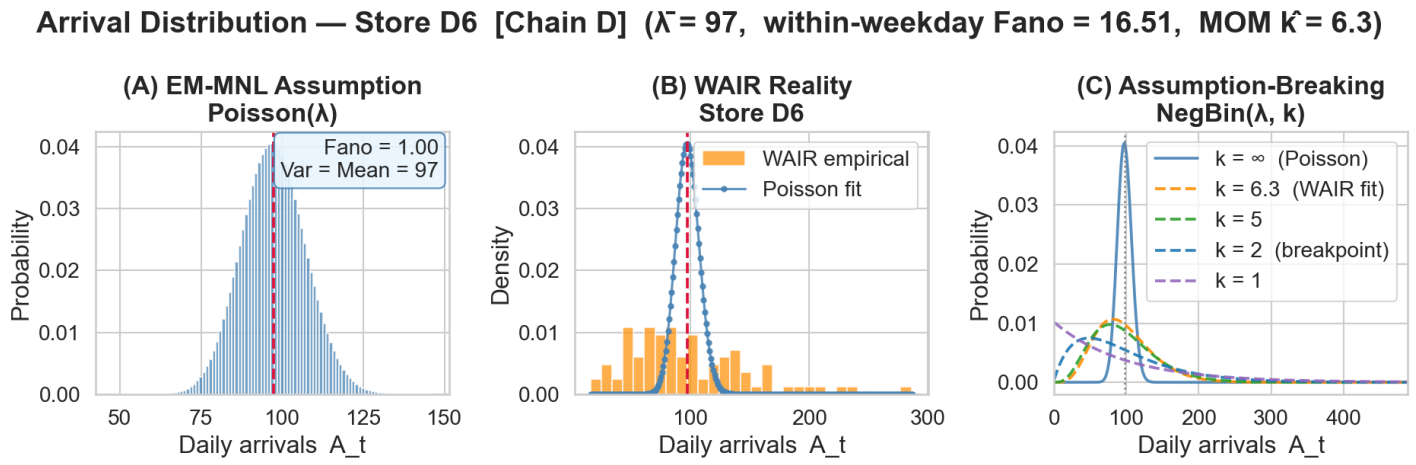


Figure D.1: Overdispersion analysis (Notebook 05). Left (A): Poisson(λ) assumption. Centre (B): empirical WAIR daily arrivals for Store D6 (Q3 2024) with a Poisson fit overlay. Right (C): the NegBin(λ, k) family at the same mean, with the Store D6 MOM estimate ($k = 6.3$) and the SQ2 breakpoint ($k = 2$) marked.

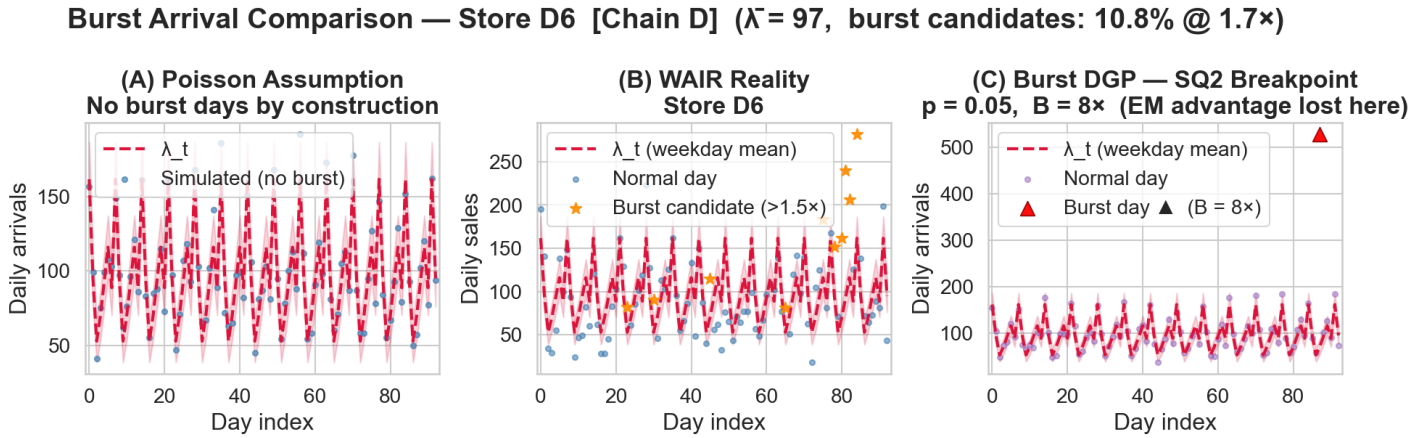


Figure D.2: Burst arrival analysis (Notebook 06), Store D6 ($\bar{\lambda} = 97$). Left (A): simulated Poisson time series (no bursts). Centre (B): WAIR empirical time series with burst candidates marked (orange stars, above $1.5\times$ the weekday mean). Right (C): simulated burst DGP at the SQ2 breakpoint ($\alpha = 8$, $p = 5\%$).

Appendix E SQ3: Real-Data Application

E.1 Applicability Gate Definitions

The SQ3 estimation sample is selected by six pre-estimation diagnostic gates, summarised in Section 3.7.2. A store is admitted to the estimation sample only if it passes all six; excluded stores are reported with the gate(s) responsible. The full definitions and thresholds are as follows.

(1) **Overdispersion gate (Fano factor).** The aggregate daily sales Fano factor $F = \mathrm{Var}(\sum_j Y_{jt})/\mathrm{E}(\sum_j Y_{jt})$ is compared to the store-specific NegBin $k=2$ breakpoint derived in Section D.1: $F_{\text{threshold}} = 1 + \bar{y}/2$, where $\bar{y}$ is the mean daily total sales. A value of $k = 2$ corresponds to the severity level at which the EM-MNL-to-Naïve MAE ratio approaches 1.0 in the SQ2 Axis 2 sweep. Stores with $F > F_{\text{threshold}}$ are excluded.

(2) **Period count.** At least $T \geq 50$ days of observations are required, consistent with the SQ1 finding that estimation bias decreases substantially beyond 50 periods [20].

(3) **Product count.** At least $J \geq 2$ SKUs must remain after filtering, as substitution is undefined for a single product.

(4) **Non-trivial censoring.** The aggregate stockout rate $1 - \bar{a}$ must exceed zero; stores where all products are in stock every day have no lost sales to recover.

(5) **Assortment variation.** The fraction of periods in which the available assortment changes must exceed zero. Without variation in S_t, the EM E-step has no variation to exploit for identifying substitution effects.

(6) **Minimum in-stock run length.** The mean length of consecutive in-stock runs per SKU must exceed 5 days. Very short runs (mean $\approx$ 1 day, as observed in the Chain B stores) indicate that products stock out almost immediately after arriving, which eliminates the extended in-stock windows required to calibrate the per-product demand rate. This gate matters because stockouts in this data are typically permanent rather than transient (Appendix A.2).

E.2 Estimation Results and Diagnostics

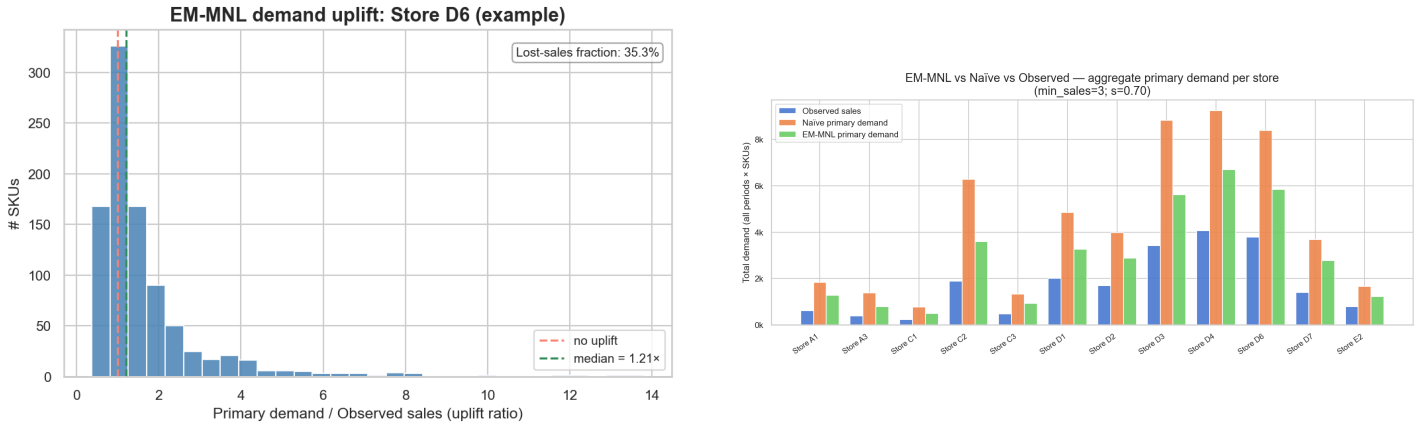


Figure E.1: Left: per-SKU demand uplift distribution for a representative store (Store D6) at $s = 0.7$. Right: aggregate EM-MNL versus naïve demand across all 12 qualifying stores.

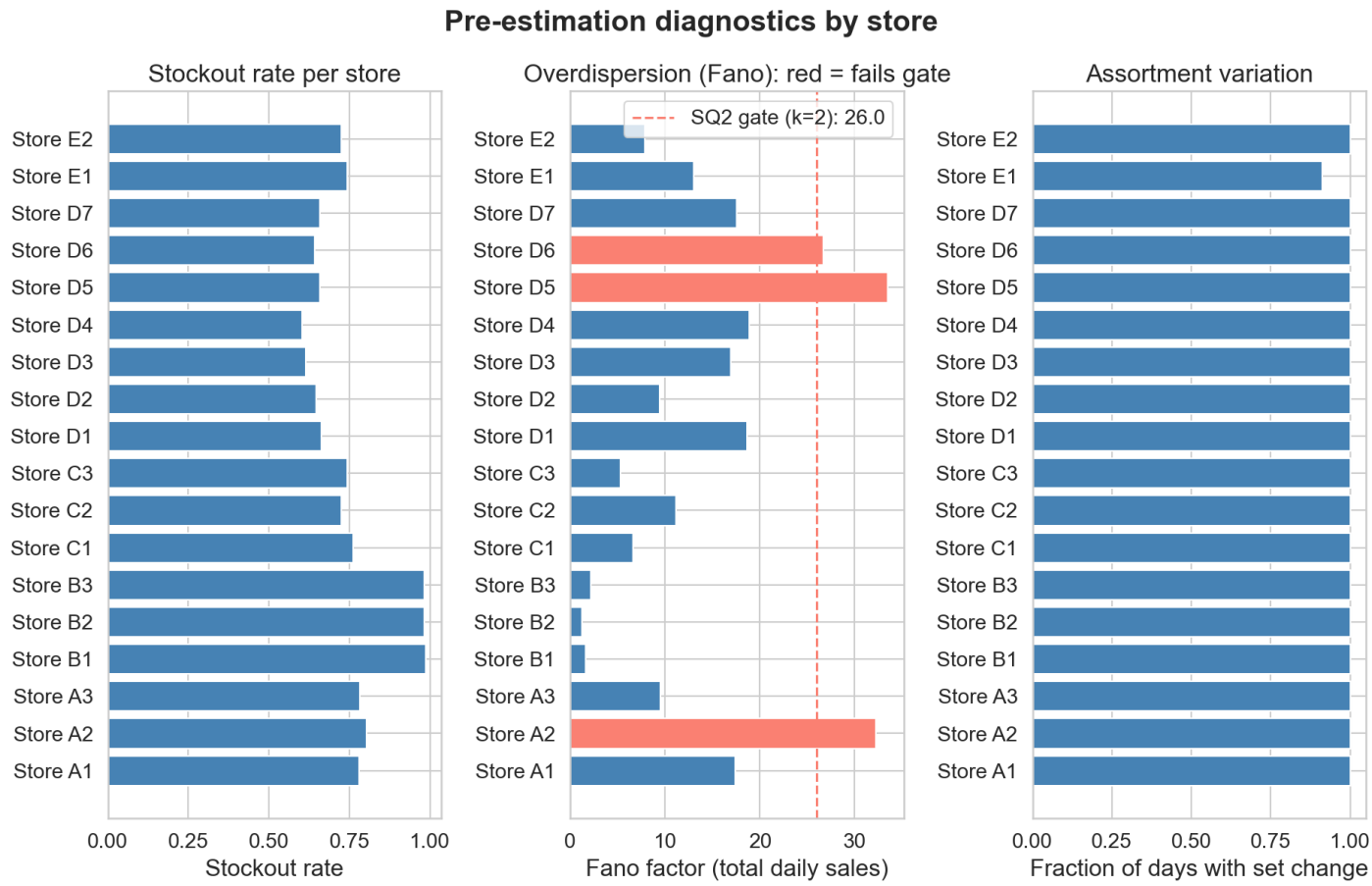


Figure E.2: SQ3 pre-estimation diagnostics for all 18 WAIR stores against the six applicability gates of Appendix E.1; stores failing one or more gates are marked.

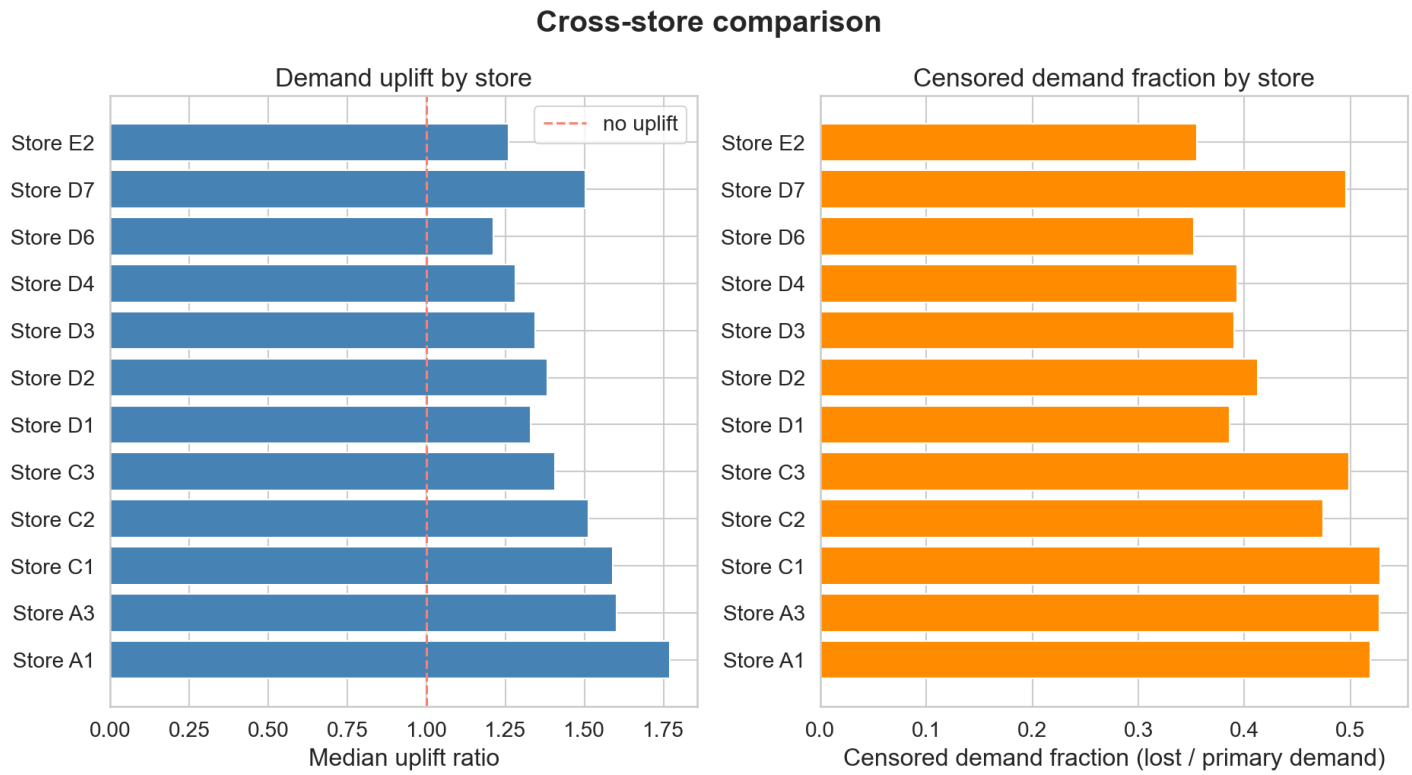


Figure E.3: SQ3 cross-store uplift consistency: per-chain comparison of median uplift distributions.

Store-level arrival rate time series. Figure E.4 shows the EM-MNL imputed daily arrival rate $\hat{\lambda}_t$ versus observed daily sales for a representative qualifying store (Store D6), illustrating the temporal demand pattern and the magnitude of the censoring correction; the remaining 11 qualifying stores show the same qualitative behaviour.

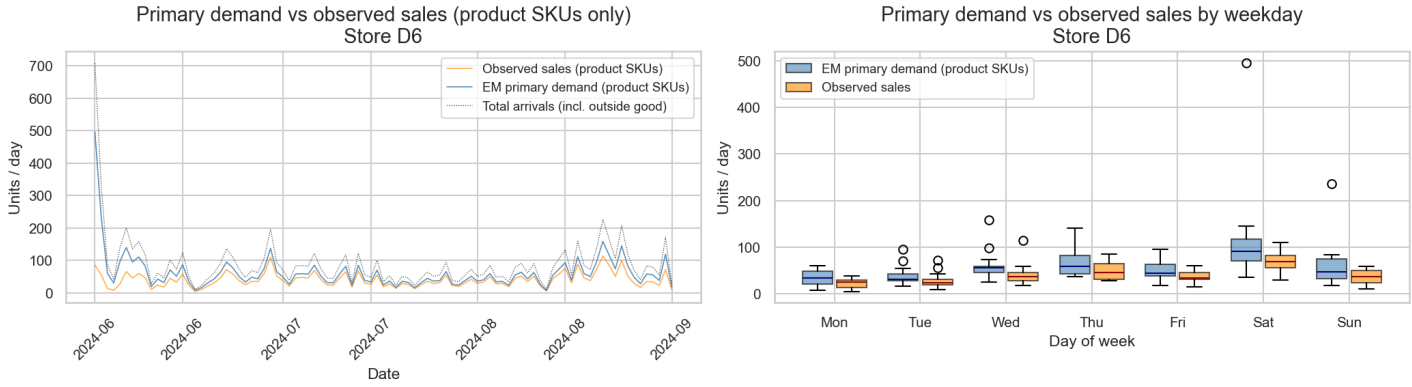


Figure E.4: Imputed arrival rates vs. observed sales for a representative store (Store D6). Left: daily EM-MNL primary demand, observed sales, and total arrivals over the observation window. Right: the same quantities aggregated by weekday.

E.3 Minimum-Sales Filter Sensitivity

Table E.1 reports the chain-level effect of the minimum-sales threshold ($s = 0.7$, all 12 qualifying stores), with the primary specification (`min_sales` = 3) in the central block. Relaxing the threshold to `min_sales` = 1 brings in the single-sale SKUs and increases median uplift to the 1.4×–9.6× range with lost-sales fractions of 0.75–0.94. These are the weakly-identified products discussed in Section 3.7.1, whose recovered demand is a high-variance extrapolation from one or two in-stock observations and cannot be assessed by the held-out evaluation. Tightening the threshold to `min_sales` = 5 reduces the sample to as few as 7–11 SKUs in Chains A and C and creates median uplift below 1× in three stores, so it serves only as an upper bound on the sweep. The qualitative findings (EM-MNL uplift exceeding the naïve baseline, cross-chain heterogeneity) are preserved at every threshold; only the magnitude of the inferred censoring correction moves with it.

[1]The `min_sales` = 3 block is the primary specification reported in Table 1 and is taken from the full per-store estimation run. The `min_sales` = 1 and `min_sales` = 5 rows are from the sensitivity sweep.

Table E.1: Minimum-sales filter sensitivity at $s = 0.7$: chain-level SKU counts, per-store median uplift ranges, and lost-sales fraction ranges at each threshold.[1]

min_sales	Chain	SKUs	Median uplift	p_{lost}
1	Chain A	6,069	7.8×–9.6×	0.93–0.94
	Chain C	4,007	2.1×–5.6×	0.83–0.94
	Chain D	18,170	1.4×–2.8×	0.75–0.90
	Chain E	1,545	4.2×	0.93
3	Chain A	286	1.6×–1.8×	0.52–0.53
	Chain C	678	1.4×–1.6×	0.47–0.53
	Chain D	3,939	1.2×–1.5×	0.35–0.50
	Chain E	159	1.3×	0.36
5	Chain A	22	1.2×–1.8×	0.53–0.56
	Chain C	138	0.9×–1.4×	0.31–0.46
	Chain D	1,080	0.9×–1.1×	0.29–0.45
	Chain E	55	0.9×	0.28

E.4 Held-Out Predictive Evaluation

Table E.2 reports per-store held-out evaluation results at the baseline masking fraction $p = 0.15$. n_{cells} is the number of masked in-stock SKU–day cells; $\Delta_{\text{dev}} = \text{Dev}_{\text{EM}} - \text{Dev}_{\text{Naïve}}$ is the mean per-cell Poisson deviance gap (negative values favour EM-MNL); the 95% CI is a percentile bootstrap interval (2,000 resamples) on Δ_{dev}; FracEMbetter is the fraction of masked cells where EM-MNL achieves lower deviance than the naïve predictor.

Table E.2: Per-store held-out predictive evaluation at masking fraction $p = 0.15$, all 12 qualifying stores.

Store	n_{cells}	MAE EM-MNL	MAE Naïve	Δ_{dev}	95% CI	FracEMbetter
Store A1	1,046	0.149	0.157	−2.51	[−7.6, 2.6]	0.631
Store A3	592	0.156	0.158	+5.76	[−5.9, 19.7]	0.606
Store C1	391	0.133	0.147	+10.51	[−0.1, 24.6]	0.621
Store C2	2,749	0.174	0.179	−4.36	[−7.3, −1.5]	0.599
Store C3	795	0.156	0.164	+13.83	[3.5, 24.2]	0.668
Store D1	3,758	0.140	0.146	−0.02	[−0.03, −0.01]	0.627
Store D2	3,131	0.143	0.147	−3.43	[−6.5, −1.1]	0.619
Store D3	5,812	0.147	0.154	−2.03	[−3.8, −0.7]	0.601
Store D4	7,594	0.145	0.150	−2.17	[−3.6, −1.0]	0.656
Store D6	7,510	0.135	0.141	−0.02	[−0.03, −0.01]	0.643
Store D7	2,613	0.141	0.147	−2.84	[−5.6, −0.5]	0.634
Store E2	1,334	0.154	0.166	−8.03	[−20.0, 1.0]	0.638

EM-MNL achieves lower MAE than the naïve predictor in every store and wins the majority of held-out cells (FracEMbetter > 0.5) everywhere. Mean deviance Δ_{dev} favours EM-MNL in 9 of 12 stores; the three positive values (Stores A3, C1, C3) occur in the smallest-assortment stores, where a few near-zero-prediction cells inflate the mean deviance.

The masking-fraction sensitivity sweep ($p \in \{0.10, 0.15, 0.20\}$) confirms that the MAE advantage and the per-cell win rate are stable across p: EM-MNL retains lower MAE than the naïve predictor in every store at every masking fraction, and FracEMbetter stays in the 0.60–0.67 range throughout.

E.5 Gate Validation: Excluded Stores

To test whether the applicability gates of Appendix E.1 are valid rather than asserted, the identical EM-MNL estimation and held-out evaluation pipeline used for the 12 qualifying stores was additionally run on the six excluded stores. This is a diagnostic exercise only: the excluded stores remain outside the estimation sample and contribute nothing to the SQ3 results or the SQ4 inputs. Table E.3 reports the outcome.

Table E.3: Gate validation on the six excluded stores

Store	Gate failed	SKUs	Run (d)	F/F_{thr}	MAE EM-MNL	MAE Naïve	Ratio	FracEMbetter
Store A2	Overdispersion	56	17.1	1.57	0.192	0.199	0.965	0.617
Store D5	Overdispersion	498	25.8	1.03	0.134	0.141	0.951	0.606
Store E1	Overdispersion	82	21.5	1.32	0.141	0.163	0.866	0.731
Store B1	Run length	0	1.0	0.29	–	–	–	–
Store B2	Run length	0	1.0	0.59	–	–	–	–
Store B3	Run length	0	1.0	0.89	–	–	–	–

The run-length gate is confirmed structurally. In all three Chain B stores the mean in-stock run is about one day, so after the min_sales = 3 filter no SKU keeps enough genuine in-stock observations to identify a preference weight. No model can be fitted and no held-out cells can be formed, so EM-MNL has nothing to work with, exactly as the gate predicts.

The held-out evaluation does not show the Fano-excluded stores failing. Across 337 to 3,877 masked cells, EM-MNL keeps a lower MAE than the naïve predictor in all three (ratio 0.87–0.97) and wins most held-out cells, placing them within the range of the 12 qualifying stores (Table E.2). However, this is not a contradiction of the exclusion. The masked-cell test measures per-SKU *sales* prediction with the arrival rate anchored, so it is blind to the arrival overdispersion the Fano gate targets. The gate instead guards against biased recovery of latent *primary demand* under overdispersed arrivals, which SQ2 established against known ground truth (Section D.1) and which no ground-truth-free real-data test can measure. The Fano gate therefore excludes these stores as a precaution calibrated to the SQ2 bias curve, not because of a visible real-data failure.

Appendix F SQ4 Neurosymbolic Extension: Formulation, Gate, and Results

This appendix gives the full formulation of the differentiable, rule-constrained estimator summarised in Section 3.8, the correctness gate that validates it, and the experiment results behind the figures and claims in Section 4.4. All the conclusions are made on simulated data where the true primary demand N_j^* is known; the real-data subsection is an illustration only.

F.1 Differentiable Estimator Formulation

Statistical model. The model is EM-MNL's. Customers arrive in period t as a Poisson process with rate λ_t and choose among the available products S_t plus a no-purchase option of weight 1 under the MNL of Equation (6). Sales of an available product j in period t are therefore Poisson with mean

$$\mu_{jt} = \lambda_t \cdot \frac{v_j}{1 + \sum_{h \in S_t} v_h}, \tag{14}$$

and the negative observed-data log-likelihood, up to an additive constant, is

$$\mathcal{L}_{\text{nll}}(\mathbf{v}, \boldsymbol{\lambda}) = \sum_{(t,j):\, a_{jt}=1} \left[\mu_{jt} - Y_{jt} \ln \mu_{jt} \right]. \tag{15}$$

λ_t is how many shoppers come in on day t, and each shopper either buys one product on the shelf or leaves with nothing. A product with a larger preference weight v_j gets picked more often. Equation (14) simply multiplies these two ideas together: the expected sales of product j equal the number of shoppers times the chance that a shopper picks j. Equation (15) is a score for how well a given set of weights and arrival rates explains the sales actually recorded; fitting the model means searching for the values that make the observed sales as likely as possible. The sum runs only over cells where the product was in stock ($a_{jt} = 1$), because a product that was out of stock could not have sold anything and so reveals nothing here.

Identification: the share anchor. The likelihood (15) does not identify $(\mathbf{v}, \boldsymbol{\lambda})$ on its own. The problem is one of scale. The data fix only how big each product's weight is *relative* to the rest of the shelf, not how big the weights are in absolute terms. We can multiply every weight by the same factor and, at the same time, nudge the arrival rates so that the predicted sales in Equation (14) do not move at all. Formally, scaling $\mathbf{v} \to c\,\mathbf{v}$ while rescaling each λ_t by $(1 + cA_t)/\big(c(1 + A_t)\big)$, with $A_t = \sum_{h \in S_t} v_h$, leaves every μ_{jt} unchanged for any $c > 0$. The two solutions fit the data equally well, so the likelihood alone cannot choose between them: it has a whole flat valley of equally good answers, known as the scale ridge.

The fix is to pin down the overall size of the weights, which removes the freedom to slide along that valley. EM-MNL does this through the market-share constraint of its M-step (Equation (9)), which holds the total weight $\sum_j v_j = s/(1 - s)$ fixed at every iteration, where s is the assumed market share. The differentiable estimator builds the identical constraint directly into how the weights are constructed,

$$\mathbf{v} = \frac{s}{1 - s}\,\text{softmax}(\boldsymbol{\theta}), \qquad \boldsymbol{\theta} \in \mathbb{R}^J \text{ unconstrained}, \tag{16}$$

where the softmax turns any list of free numbers $\boldsymbol{\theta}$ into positive shares that add up to one, and the leading factor scales those shares to the required total. Because that total is fixed by construction, the scale ambiguity simply cannot arise, and running ordinary gradient descent on $\boldsymbol{\theta}$ solves exactly the same scale-constrained problem EM-MNL addresses.

Rule penalty. The idea behind the rule is simple: products that belong to the same style, for example the same shirt in different sizes, should have similar preference weights, so when the data are thin on one sibling the others can lend it support. A logical implication over a continuous quantity cannot be scored by the exact semantic loss of Xu et al. [24], which is defined for Boolean constraints through weighted model counting. Following the semantic-based-regularisation tradition [2, 4], the rule of Equation (4) is relaxed into a penalty that is zero when the rule holds exactly and grows with the size of the violation:

$$\mathcal{L}_{\text{rule}} = \sum_{c} \sum_{\substack{i,j \in C_c \\ i<j}} (v_i - v_j)^2, \tag{17}$$

where C_c is the set of SKUs in style group c. In words, Equation (17) adds up the squared gap between every pair of same-style siblings: it is zero when the siblings have equal weights and grows the further apart they drift, so including it in the objective gently pulls sibling weights together. This is a graph-smoothness penalty over same-style edges; a style with only one SKU has no pairs and so adds nothing. The full objective is $\mathcal{L} = \mathcal{L}_{\text{nll}} + w \cdot \mathcal{L}_{\text{rule}}$ (Equation (5)), where the single weight w is the dial that sets how hard the rule pulls: $w = 0$ ignores it entirely and larger w enforces it more strongly.

Reported demand and optimisation. Finally, the fitted weights are turned back into a demand number. Equation (18) answers the question: if every product had been in stock for the whole window, how many units of product j would customers have wanted? This full-assortment expectation strips out the effect of stockouts and so is directly comparable across estimators and to the simulated truth,

$$\hat{N}_j = \Big(\sum_t \hat{\lambda}_t \Big) \frac{\hat{v}_j}{1 + \sum_i \hat{v}_i}. \tag{18}$$

The objective is minimised in two passes: Adam first takes a fast run down towards the optimum, and L-BFGS then polishes the final answer, so the optimum is resolved tightly enough for the correctness gate below.

F.2 Correctness Gate

At $w = 0$ the rule is switched off, so the estimator is solving exactly the same scale-anchored likelihood that EM-MNL solves. The natural sanity check is therefore that the two should arrive at the same answer. The original gate phrased this as "match EM-MNL's per-product demand within 1%", and it proved impossible to pass, for an informative reason: EM-MNL itself does not quite reach the best-fitting solution. The point its iterations settle on (its fixed point) is not the exact scale-constrained maximum-likelihood estimate (MLE). Two separate checks point the same way. First, letting EM-MNL run essentially forever, until its steps shrink to a step norm of 10^{-12}, still leaves it at a point that fits the data slightly worse than the gradient solution. Second, a completely independent third implementation (SciPy L-BFGS-B on the profile likelihood, which removes λ from

the problem analytically by substituting $\lambda_t = m_t(1 + A_t)/A_t$ with $m_t = \sum_{j \in S_t} Y_{jt}$) agrees with the gradient solution to seven decimal places, and not with EM-MNL. In other words, it is the gradient method that lands on the true optimum, while EM-MNL stops just short of it. The cause is EM-MNL's ratio-scaling E-step (Equations 7–8), an approximation that is exact only when nothing is out of stock and whose small error grows as more products are censored.

Because EM-MNL is therefore not a gold standard to be copied, the check is rebuilt around the independent MLE instead. The gate is strengthened, not relaxed, into three hard assertions; all pass on the rule-true test bed at ≈ 50% censoring:

(1) **Exactness.** The differentiable estimator's $\hat{N}_j$ matches the independent SciPy MLE within 0.1% per product (observed maximum gap 4.6×10^{-7}, ten times inside the original tolerance).
(2) **Never worse.** Its negative log-likelihood is no higher than EM-MNL's on the shared objective (observed gap −0.079).
(3) **Demand-recovery equivalence.** Its MAPB (5.97%) is within 0.5 percentage points of EM-MNL's (6.19%).

Table F.1 measures how far EM-MNL drifts from the exact MLE as censoring increases. The gradient estimator stays level with the independent MLE at every level, whereas EM-MNL's per-product demand gap grows from about 0.7% at 10% censoring to about 1.9% at 50%. The drift comes from a small period-varying factor $(A_t + 1)$, where A_t is the total weight of the in-stock products in period t, that EM-MNL's converged weights carry but the exact MLE does not: it has no effect when nothing is out of stock and grows as more stock runs out. The drift is small, so EM-MNL remains an excellent approximation in practice. The MM algorithm of Abdallah and Vulcano [1] converges to the exact constrained MLE and does not exhibit it.

Table F.1: EM-MNL versus the exact scale-constrained MLE across censoring levels (rule-true test bed, $J = 10$, $T = 100$). "Max gap" is the maximum per-product relative difference in primary demand $\hat{N}_j$ against the independent SciPy MLE; $\Delta\mathcal{L}_{\text{nll}}$ is the EM-MNL negative log-likelihood minus the MLE's (positive means EM-MNL is worse).

p_{avail}	Censoring	EM-MNL max gap	DiffMNL max gap	$\Delta\mathcal{L}_{\text{nll}}$
0.9	10.4%	0.70%	9.0×10^{-8}	0.030
0.8	19.8%	1.59%	1.9×10^{-7}	0.061
0.7	28.6%	1.24%	1.7×10^{-7}	0.091
0.5	50.8%	1.85%	4.6×10^{-7}	0.079

F.3 Rule-Weight Sweep (Rule True)

This subsection increases the rule weight w on data where the rule is true and reports the findings. The weight is swept over $w \in \$\{0, 1, 3, 10, 30, 100, 300, 1000, 3000\}$, and three things are tracked at each setting: how close the recovered demand is to the truth (MAPB against N_j^*); how well the model predicts sales it was not trained on (held-out mean Poisson deviance, using the SQ3 masked-cell protocol of Section 3.7.3, averaged over three mask seeds, with the SQ3 mean-floor of 10^{-3} applied to every method); and how closely the rule is obeyed (the mean spread of $\hat{v}$ within a style, which shrinks as same-style weights are pulled together). Figure F.1 and Table F.2 report all three. The picture is consistent: demand recovery improves steadily as w rises, dropping below the EM-MNL reference once $w \gtrsim 100$ and reaching 3.57% at $w = 3000$; out-of-sample prediction is best near $w = 1000$; and the within-style spread falls steadily towards zero. That last curve is the operational point, because rule adherence is something we can actually measure, and w is the dial that sets it.

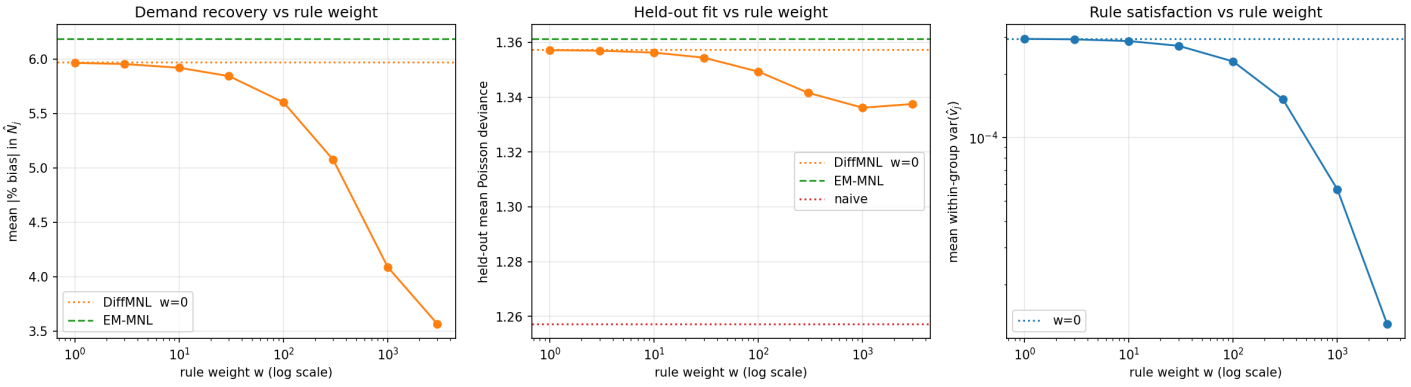


Figure F.1: SQ4 rule-weight sweep on rule-true simulated data ($J = 10$, $T = 100$, ≈ 50% censoring). Left: demand-recovery bias (MAPB) against the known N_j^*. Centre: held-out mean Poisson deviance. Right: rule satisfaction (mean within-style variance of $\hat{v}$).

One caveat is the naive per-SKU mean attains a lower raw held-out deviance (1.257) than any model-based method in this simulated setting. With availability drawn independently per cell, masked cells are exchangeable with unmasked ones, so a pooled per-SKU mean is a strong predictor, whereas the model-based predictions inherit noise from each period's individually estimated $\hat{\lambda}_t$. The ordering reverses on both real stores (Appendix F.7), where stockouts are persistent and assortment conditioning matters. The naive mean predicts observed sales but has no concept of primary demand, so it cannot address the SQ1/SQ3 estimation problem at all; the decisive column remains bias against the known N_j^*.

Table F.2: SQ4 rule-weight sweep on rule-true data

w	MAPB	Within-style var($\hat{v}$)	Held-out deviance
0	5.97%	3.0×10^{-4}	1.3573
1	5.96%	3.0×10^{-4}	1.3572
3	5.95%	3.0×10^{-4}	1.3570
10	5.92%	3.0×10^{-4}	1.3563
30	5.84%	3.0×10^{-4}	1.3545
100	5.60%	2.0×10^{-4}	1.3494
300	5.08%	2.0×10^{-4}	1.3417
1000	4.09%	1.0×10^{-4}	1.3362
3000	3.57%	≈ 0	1.3375

F.4 Rule-Weight Bound and Spread Sensitivity

Because w is set by hand, the practical question is how far it can be raised before it stops helping, which depends on how true the rule is. The sweep is extended to $w = 10^5$ at three within-style spreads $\sigma \in \{0.05, 0.10, 0.20\}$, where a larger σ is a looser rule. From each curve we read w^*, the weight giving the lowest bias, and w_{upper}, the first weight past w^* where bias returns to the EM-MNL baseline. Each spread is a fresh dataset, so each curve is read against its own baseline, not across spreads. Figure F.2 and Table F.3 report the result.

At $\sigma = 0.05$ neither bound exists in range: the curve keeps falling to a 3.39% asymptote (EM-MNL 6.19%), because forcing equality barely distorts a near-uniform group. Once siblings genuinely differ, a bound appears. At both $\sigma = 0.10$ and $\sigma = 0.20$ the optimum sits at $w^* \approx 700$, each below its EM-MNL baseline (4.73% vs 5.51%, and 4.74% vs 7.30%), but the useful band narrows as the rule loosens: w_{upper} falls from $\approx 10^4$ to $\approx 5 \times 10^3$. A safe default is therefore a weight of order 10^2 to 10^3, below w_{upper} for any plausible spread; on real data the rising held-out deviance of Appendix F.3 is the external guard.

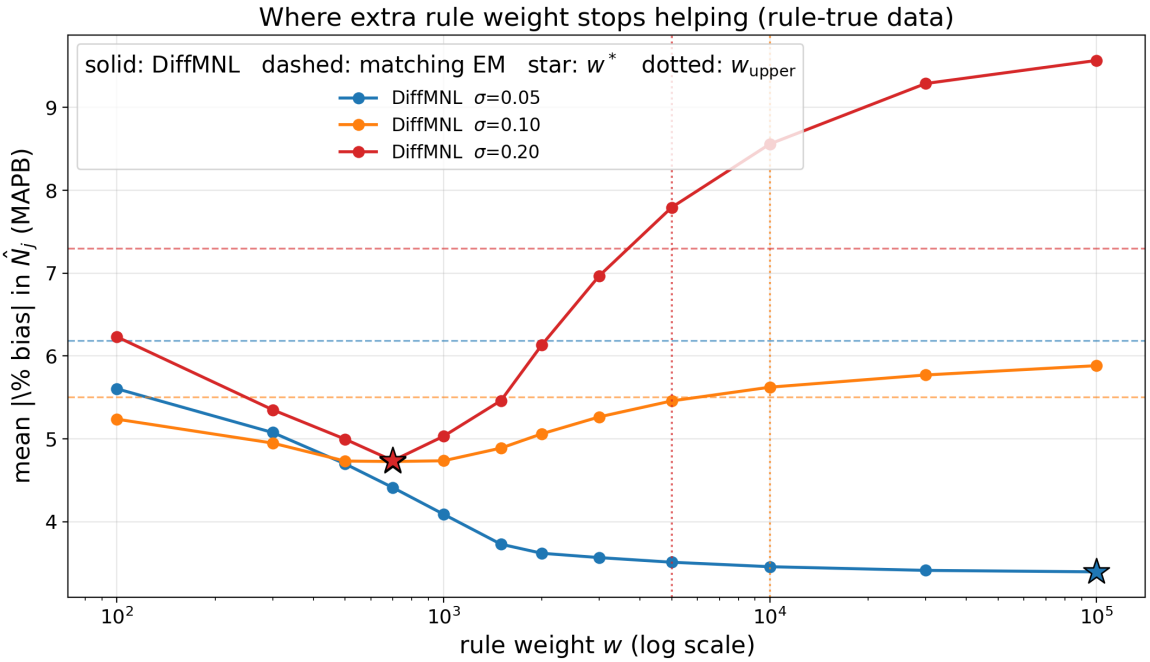


Figure F.2: SQ4 rule-weight bound on rule-true data ($J = 10$, $T = 100$, $\approx 50\%$ censoring), shown at three within-style spreads σ. The optimum weight w^* and the upper useful weight w_{upper} are marked on each curve.

Table F.3: SQ4 rule-weight bound across within-style spreads σ. Each row is a separate simulated dataset, so levels are read against the row's own EM-MNL baseline, not across rows. w^* is the optimum (the weight minimising MAPB), min MAPB the bias at w^*, and w_{upper} the first weight past w^* at which MAPB rises back above EM-MNL.

σ	EM-MNL MAPB	w^*	min MAPB	w_{upper}
0.05	6.19%	($\to 10^5$, asymptote)	3.39%	none in range
0.10	5.51%	≈ 700	4.73%	$\approx 10^4$
0.20	7.30%	≈ 700	4.74%	$\approx 5 \times 10^3$

F.5 Sample Efficiency

The prior interpretation of w predicts that the rule helps most when the data are weakest: with few periods the per-SKU evidence is noisy and borrowing strength across a style group should pay, while with many periods the likelihood dominates and the rule fades to harmless. Figure F.3 and Table F.4 test this over horizons $T \in \{15, 30, 60, 100\}$, five fresh simulations per horizon. At $T = 15$ the constrained estimator roughly cuts the demand-recovery bias of unconstrained estimation by a third (from EM-MNL's 14.6% to 9.3% at $w = 1000$); the absolute gain narrows as T grows. Borrowing strength within style groups substitutes for data the censored panel does not contain.

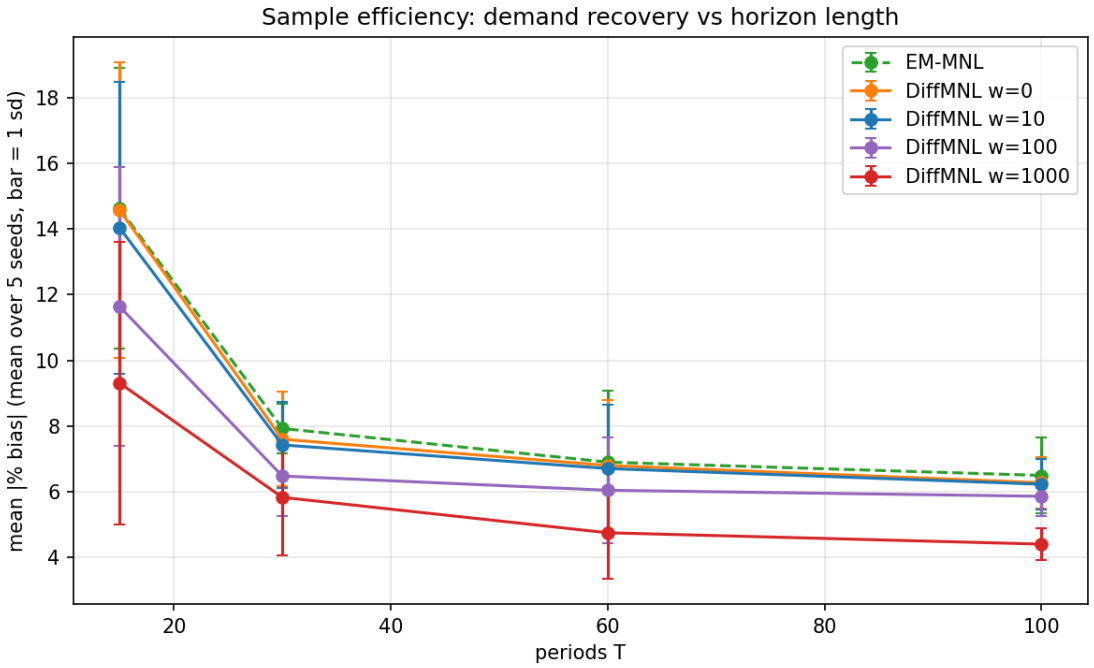


Figure F.3: SQ4 sample efficiency: demand-recovery bias (MAPB against N_j^*) versus horizon length T, mean over five simulations per horizon (error bars one standard deviation).

Table F.4: SQ4 sample efficiency: mean MAPB (%) over five simulations per horizon, for EM-MNL and the differentiable estimator at four rule weights. The advantage of a non-zero weight is largest at $T = 15$ and narrows as the horizon grows.

T	EM-MNL	w = 0	w = 10	w = 100	w = 1000
15	14.63	14.58	14.04	11.64	**9.31**
30	7.92	7.60	7.42	6.47	**5.83**
60	6.90	6.80	6.70	6.04	**4.75**
100	6.49	6.27	6.22	5.86	**4.40**

F.6 Stress Test: a False Rule

To confirm the penalty is working, the same sweep is run on data with no style structure (preference weights drawn independently) while the penalty keeps the style labels, so the rule asserts a similarity that is absent. Figure F.4 and Table F.5 report the result over three fresh simulations. At $w = 0$ the estimator matches EM-MNL (5.7% versus 5.4% MAPB), as it must; degradation is then monotone in w, reaching 18.8% at $w = 1000$ and 28.7% at $w = 3000$ towards the hard-constraint limit, where all group members are forced to a common value. At the weights where a true rule helped most, a false rule multiplies the bias. This gives two key takeaways: a constraint that could not worsen performance when wrong could not help when right, and the estimator gives no internal warning that the encoded knowledge is wrong, so its quality is the practitioner's responsibility.

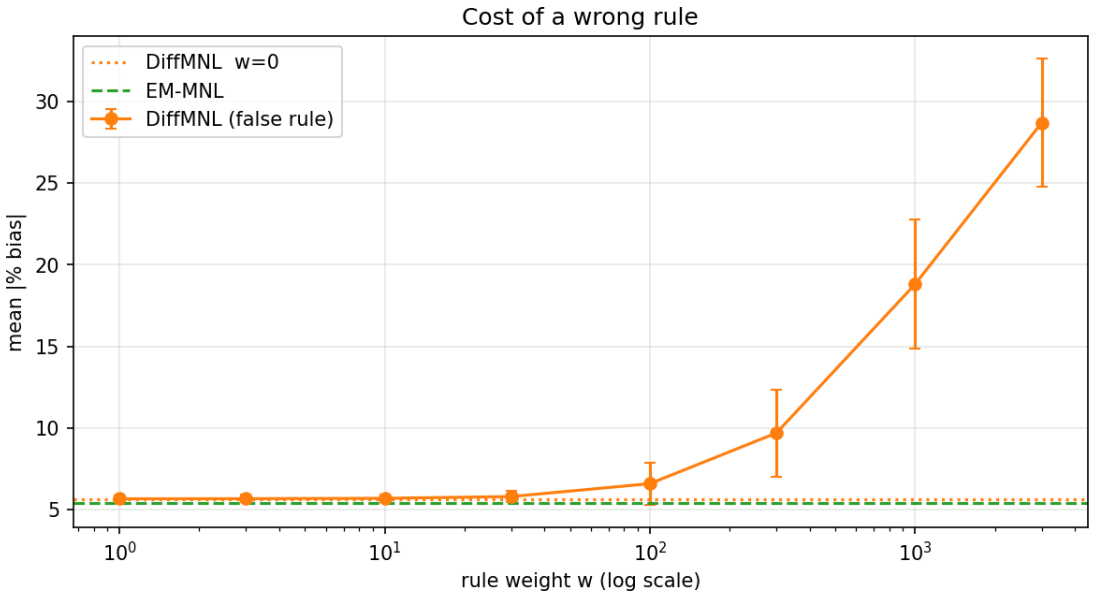


Figure F.4: SQ4 stress test: demand-recovery bias (MAPB against N_j^*) when the style rule is false in the data-generating process but the penalty is applied regardless, over three fresh simulations (error bars one standard deviation).

Table F.5: SQ4 stress test: mean MAPB (%) over three fresh simulations with the style rule false in the data-generating process.

w	0	1	3	10	30	100	300	1000
MAPB	5.66	5.66	5.67	5.69	5.80	6.59	9.69	18.81

At $w = 3000$ the bias reaches 28.69%.

F.7 Real-Data Illustration

The method is finally tested on two real qualifying SQ3 stores, grouping individual products by their style (`style_id`). Any style containing only a single product is ignored in the penalty calculation, since it has no siblings to be pulled towards. Because the real world has no perfect answer key for true demand, success is measured by how well each model predicts new, unseen sales (held-out mean Poisson deviance, Table F.6).

In both stores the new differentiable method beats the older standard EM-MNL method out of sample, and both beat a basic naive guessing baseline. This reverses the simulated ordering and confirms that using a smarter, model-based estimator really pays off when dealing with messy, real-world data.

Turning up the strength of the rule (setting the weight to 1000) moves the held-out predictions slightly but consistently in the right direction, while making the estimates for products within the same style much more consistent (roughly halving the within-style variance of $\hat{v}$). No massive breakthrough in accuracy is claimed: the improvements look small because the sheer amount of data. The main point is simply to show that the method runs on massive, real-world datasets, and that the rule can be tuned on real data without hurting held-out fit.

Table F.6: SQ4 real-data illustration on two qualifying SQ3 stores at $s = 0.7$, $T = 93$

Store	**SKUs**	**Multi-SKU styles**	**Naive**	**EM-MNL**	**DiffMNL w=0**	**DiffMNL w=1000**
Store E2	159	38 (51%)	0.5903	0.5572	0.5547	0.5540
Store D1	499	81 (39%)	0.4345	0.4137	0.4127	0.4123

Appendix G Generative AI Usage Statement

I used the AI as a support tool while researching and writing this thesis.

I used it for code development. The AI helped me write, review, and debug Python code using Polars, NumPy, scikit learn, and PyTorch for the simulation studies, the EM-MNL estimator, and the DiffMNL extension. I made all the modeling decisions myself. I validated and thoroughly checked all the code before relying on its output.

I used the AI for formatting too. It helped me get the document into the ACM template, organize the structure, and manage my citations.

For language editing, I wrote the text myself and used the AI to improve the grammar, phrasing and punctuation to make my writing clearer. I also used it to help organize sections and check my drafts against the assessment guidelines.

The AI did not generate the research questions, the methodology, or the core contributions of this project. The ideas for the EM-MNL reformulation came from research and discussions with my supervisors. The analysis of the divergence between the EM-MNL fixed point and the constrained maximum likelihood estimator is my own work which I used AI generated code to validate. The design of the DiffMNL reformulation with its substitution penalty and correctness gate is also my own work. I used the AI as a sounding board for the DiffMNL to help validate and correct the maths as I worked on it. I did all the experimental work on the WAIR dataset myself. The AI did not produce any results, data, or analyses. I verified all sources myself and did not use AI to generate citations. I reviewed all the AI outputs and take full responsibility for the content of this thesis.